

CAPSTONE PROJECT 2
Final Report

Telco Churn Analysis: A review of Sampling Techniques

by

Ivan Sing Wen Sie
21063698

Bachelor of Information Systems (Data Analytics)

Supervisor: Prof Selina Low Yeh Ching

Semester: September 2024

Date: 19 December, 2025

School of Computing and Artificial Intelligence
Faculty of Engineering and Technology
Sunway University

Abstract

The issue of customer churn prediction is extremely important in the telecommunications sector because it is much cheaper to retain a customer than bring a new one. Nevertheless, churn datasets are usually defined by extreme class imbalance with churners being a minority group, and many machine learning algorithms tend to predict non-churn customers biasedly. The research paper examines how different methods of data sampling can influence the performance of the churn prediction using a telecom customer dataset in the public. It compares four commonly used classification models Decision Tree, Logistic Regression, Random Forest, and Support Vector Machine (SVM) on various resampling strategies, such as SMOTE, SMOTE-Tomek, SMOTE-ENN and ADASYN. CRISP-DM framework is embraced to inform the process of data understanding, preparation, modeling and evaluation. The metrics are model performance against metrics appropriate to the imbalanced classification and especially F1-score and Area Under the ROC Curve (AUC) with a focus on minority class. The outcome of the experiments shows that the choice of a sampling strategy can significantly affect the effectiveness of the model. Hybrid sampling algorithms, in particular SMOTE-ENN, provide the most significant gains in tree-based models, whereas Logistic Regression is stable and robust in terms of performance, reaching the highest results with Cluster-SMOTE. SVM best utilizes the conventional SMOTE-based oversampling because it forms better margins. Conversely, the performance of all models is harmed by aggressive undersampling techniques like NearMiss. Moreover, the methods of generative oversampling can be not very effective on a tabular telecom data. The results point out that proper sampling technique, and classical models of machine learning combinations can greatly increase the accuracy and reliability of churn predictions, which could provide a useful recommendation to implement an interpretable and efficient churn prediction system into practice in a telecom settings.

Contents

1.0 Introduction.....	3
1.1 Problem Statement.....	4
1.2 Project Objective.....	5
1.3 Scope.....	6
2.0 Literature Review.....	6
2.1 Predictive Models for Churn Detection and Justification.....	7
2.1.2 Handling Class Imbalance in Churn Data.....	10
2.1.2.1 Oversampling Techniques.....	12
2.1.2.2 Undersampling and Hybrid Cleaning Techniques.....	13

2.1.3 Interpretability and Explainability	14
2.1.4 Operational and Business Considerations	15
2.1.5 Summary	17
3.0 Methodology	19
3.1 CRISP-DM	19
3.2 Business Understanding	21
3.3 Data Understanding	21
3.4 Data Preparation	23
3.5 Model Development and Resampling Techniques for imbalance	24
3.5.1 Synthetic Minority Over-sampling Technique (SMOTE)	25
3.5.2 SMOTE with Tomek Links (SMOTE-Tomek)	26
3.5.3 SMOTE + Edited Nearest Neighbours (SMOTE-ENN)	28
3.5.4 Adaptive Synthetic Oversampling (ADASYN)	29
3.6 Classification Models	30
3.6.1 Decision Tree Classifier	30
3.6.2 Logistic Regression	32
3.6.3 Random Forest	33
3.6.4 Support Vector Machine (SVM)	35
3.7 Evaluation	38
3.8 Tools, Technology and Deployment	40
4.0 Results	41
4.1 Overall Performance Summary	41
4.2 F1-AUC Trade-off Patterns	45
4.3 Key Observations from Results	45
5.0 Discussion	46
5.1 Impact of Sampling techniques	46
5.2 Performance of Tree-Based Models	46
5.3 Performance of Logistic Regression	48
5.4 Performance of Support Vector Machine	48
5.5 Insights from the F1-AUC Scatter Plot	49
5.6 Practical Implications for Churn Prediction	49
5.7 Summary of Discussion	50
6.0 Conclusion	50
6.1 Overall Conclusion	50
6.2 Limitations	52

6.2.1 Data Constraints	52
6.2.2 Model Scope	52
6.2.3 Sampling Method Constraints	52
6.2.4 Hyperparameter and Pipeline Limitations	52
6.2.5 Evaluation Metric Limitations	52
6.3 Future Work	53
6.3.1 Expanded model Exploration	53
6.3.2 Improved Generative Oversampling Methods	53
6.3.3 Cost-Sensitive and Ensemble-Based Approaches	53
6.3.4 Hyperparameter Optimization and Sampling Intensity Tuning	53
6.3.5 Integration of Explainability and Business Metrics	53
6.3.6 Use of temporal and Behavioral Data	53

1.0 Introduction

In today's highly competitive telecommunication industry, companies are struggling just to retain their customers on a daily basis and has become a crucial part of the business as compared to acquiring new ones. This is due to the increasing numbers of service providers and the ease of switching between them, customer churn is the percentage of customers who stopped using your company's product or service during a certain timeframe. These churned customers may leave for a competitor or stop using your offering without an alternative [1]. According to studies, the cost of acquiring new customers can be five to six times higher than the cost of trying to retain their existing customers. As such, accurately identifying customers who are at risk of churning and understanding their reasons behind it is crucial for profitability.

With models that can identify patterns in past customer behaviour to forecast attrition or any other patterns and trends in other settings, machine learning has been widely used for applications such as churn prediction. However, a major challenge in building reliable predictive models for churn is the class imbalance typically found in such datasets. Class imbalance occurs when one class in a classification problem significantly outweighs the other class. A model trained on an imbalanced dataset perform poorly on the minority class. At best, this can cause loss to the business in the case of a churn analysis. Churners often represent only a small portion of the data, causing many machine learning models to become biased toward the majority class which are people who did not churn, which results in poor predictive performance on the minority class.

This project will focus on analyzing telecom churn data using multiple machine learning algorithms with a particular focus on different types of sampling strategies to address data imbalance. The project will be evaluating traditional and advanced resampling techniques such as SMOTE, SMOTE-Tomek, SMOTE-ENN, and ADASYN to determine their impact on churn prediction performance across various classifiers. By improving the model's ability to detect churners, this research aims to support data-driven retention strategies that telecom companies can adopt.

1.1 Problem Statement

Despite the increasing use of data analytics in the telecom sector, many companies today still face challenges in accurately predicting customer churn. Traditional models often fail to account for data imbalances, evolving customer behavior, and the complex interactions between variables because of how vast the data is. As a result, predictions may be unreliable, leading to ineffective or delayed retention actions.

Class imbalance is one of the most significant issues that telecommunication firms confront when it comes to effectively forecasting customer attrition. Customers who leave are much less common than those who remain in most real-world telecom datasets. Many machine learning models become biased because of this mismatch, favouring the majority class and missing churners, which are the exact group that companies need to find.

While there are traditional sampling methods like SMOTE and random undersampling are the ore commonly used methods to address these problems, they are not always effective. Problems such as the loss of information and overfitting will occur. There exists a growing need to explore more advanced and hybrid sampling techniques that would better preserve class boundaries, handle noise, and also improve prediction metrics such as recall and F1-score for the minority class.

Therefore, this study investigates and compare the effectiveness of several sampling strategies in the context of churn prediction. The goal is to identify methods that can significantly improve model performance for churn detection without sacrificing generalization capability.

1.2 Project Objective

This project aims to address the critical issue of class imbalances in churn analysis and to improve the accuracy and its reliability of customer churn prediction in the telco sector. Specifically, the study focuses on evaluating and comparing the effectiveness of various sampling strategies—including SMOTE, ADASYN, SMOTE-Tomek, and SMOTE-ENN—in enhancing the performance of churn classification models.

This research will create predictive models by using resampled datasets and historical telecom customer data to evaluate the impact of each technique on churning detection. Data preprocessing, exploratory data analysis (EDA), and the use of several machine learning classifiers, including ensemble models, logistic regression, and decision trees, will all be part of the procedure. Metrics including accuracy, precision, recall, F1-score, and AUC-ROC will also be used to assess each of the model's performance, with an emphasis on enhancing the minority class recognition.

The final delivery of this project will include a comparative analysis of the sampling methods, highlighting which of the techniques used is the most effective at balancing out the dataset and improving its predictive performance. Insights drawn from the evaluation will help telecom companies implement more proactive and data-driven customer retention strategies.

- To analyze customer churn data from the telecom sector with an emphasis on class distribution and its impact on model performance.
- To implement and compare various sampling techniques, including SMOTE, SMOTE-Tomek Links, SMOTE-ENN, and ADASYN, for handling class imbalance in churn datasets.
- To evaluate the impact of different sampling techniques on the performance of multiple machine learning models, including Decision Tree, Logistic Regression, Random Forest, and Support Vector Machine (SVM), using evaluation metrics such as F1-Score, and Area Under the ROC Curve (AUC), with particular emphasis on the minority churn class.

1.3 Scope

This project looks into customer churn in the telecom industry, with a main focus on how to deal with class imbalance in the dataset. In many real-world cases, there are way more

customers who stay than those who leave, and that makes it harder for machine learning models to correctly identify the ones who are likely to churn. So, the goal here is to test out and compare different sampling methods to see which ones actually help improve the model's performance.

The dataset I'm using is a public telecom dataset that I have found from Kaggle which includes details like customer demographics, billing info, and service subscriptions. After preparing and cleaning the data, I'll apply several sampling techniques like SMOTE, ADASYN, SMOTE-Tomek, and SMOTE-ENN to balance it out. Then, I'll use different machine learning models—including logistic regression, decision tree, random forest, and SVM—to see how each sampling method affects the results.

I'll evaluate the models using common metrics like accuracy, precision, recall, F1-score, and AUC. Everything will be done in Python using libraries like scikit-learn and imbalanced-learn. While this project doesn't involve deploying a real-time system, it should still give some useful insights into which sampling strategies work best for predicting customer churn more accurately.

2.0 Literature Review

Customer churn prediction has become an essential area of research in the telecommunications industry, it is driven by the growing need to retain customers in an increasingly competitive market. As organizations generate vast amounts of customer data, machine learning and statistical modeling techniques have been widely adopted to identify patterns that signal customer disengagement. This literature review explores key developments in churn prediction, focusing on predictive modeling approaches, strategies to handle data imbalance, and methods to enhance model interpretability. Subsequent sections will examine existing models, evaluate sampling techniques, and provide a comparative analysis of classifiers commonly used in churn detection.

2.1 Predictive Models for Churn Detection and Justification

Ullah et al. [2] proposed a comprehensive churn prediction model using Random Forest, which achieved an accuracy rate of 88.63%. Their approach integrated classification with clustering techniques and employed feature selection methods such as information gain and correlation attribute ranking. The model was then validated using a real-world dataset, and k-means clustering was also applied for customer profiling. While Random Forest produced a high

accuracy outcome, its interpretability remained a challenge, prompting them to use attribute-selected classifiers to extract decision rules.

In another study, Dahiya and Bhatia [3] examined the effectiveness of both Decision Trees and Logistic Regression for churn classification. Their findings showed that the J48 Decision Tree has outperformed Logistic Regression in terms of accuracy across small, medium, and large datasets. The study has highlighted the importance of using decision rules for churn classification, but they did not delve into the feature selection or data imbalance issues.

Other studies like the one by Ahmed and Maheswari [4] have supported the use of ensemble models in churn predictions. They did so by reviewing various machine learning and metaheuristic algorithms, this included Particle Swarm Optimization (PSO), and concluded that hybrid models that combines sampling, feature selection, and classifier ensembles deliver a much better performance than usual standalone methods. These findings have pushed the field toward integrating multiple techniques for optimal accuracy.

Not only that, another article by Bhatnagar and Srivastava [5] has evaluated numerous algorithms such as Decision Tree, Logistic Regression, SVM, Naive Bayes, KNN, XGBoost, etc. on a telecom churn dataset and found that a Random Forest classifier performed “exceptionally well” with an approximate result of 90% accuracy, it is the best among the tested models [5]. On the other hand, another similar report by Poudel et al. [6] states that a Gradient Boosting Machine (GBM) model was the top performer with an estimate accuracy of 81% in their churn testing, it comes with statistical tests that confirms its superiority over the other classifiers [6]. These results underscore that tree ensembles often top the accuracy rankings for churn prediction.

However, the gap between these advanced models and traditional classifiers are not always dramatic. Some of the research highlights that simpler algorithms can actually result in comparable performances but only under certain conditions. The article by Akbar and Apriono [7] showed that while XGBoost achieved the highest accuracy of ~81.6 on a telecom churn task, a basic Decision Tree and even a Naïve Bayes model is able to produce a similar outcome with an AUC of (**≈0.75**) as XGBoost [7]. In other terms, usually the more interpretable models kept pace in terms of ranking performances (AUC), despite a slight accuracy deficit. In a related finding, Prabadevi et al. [8] has observed that a deep learning model like neural network has

edged out other models for the highest accuracy on a Kaggle telecom churn dataset, but yet a Logistic Regression model was deemed to be the “most interpretable and reliable” despite its lower accuracy [9].

There is previous research that has emphasised that greater reduction in prediction error does not necessarily lead to improved churn classification performance. Random Forest and XGBoost are examples of ensemble-based algorithms, which are often reported to have low error values because of their capability to capture complex nonlinear association; nevertheless, a number of studies warn against the use of boosted models because they can have overfitting or unstable generalization when tested on telecom churn datasets with imbalances; especially when aggregate error measures are used as the evaluation metric [9]. This drawback has cast doubts about their usefulness in practical decision-making processes, in which it is important to identify churners consistently.

Conversely, Random Forest has been highly known as a strong and stable classifier with a balance between predictive accuracy and over-fit. As indicated by existing literature, Random Forest has always been shown to be effective in terms of a variety of evaluation metrics and has a high generalisation capacity on unknown data, making it a frequent choice in studies on telecom churn prediction [9]. Its ensemble format also increases strength due to the minimization of variation as compared to single-tree models.

Even now, Logistic Regression remains a robust base model in the studies of churn prediction because of its interpretability and reliability. Some of the studies underline that Logistic Regression can be used to achieve competitive accuracy and predictable results in most cases, especially when the data is properly pre-processed and feature engineering is used [8], [9]. Its explainability and openness are particularly effective when used in the business setting, since it is not only necessary to know the factors that drive the customer churn but also the predictability level.

Despite the potential of deep learning and neural network-based methods in modelling complex patterns, previous studies show varying findings in the activity of churn prediction. These models generally come at a greater computational price and are susceptible to overfitting particularly in situations where training data is scarce or lopsided [9]. As a result, most of the researches prefer the classic machine learning algorithms like Decision Tree, logistic

regression, Support Vector Machine (SVM), and Random Forest to predict churn-based studies because of their predictability, understandability, and availability of easy deployment.

To substantiate this fact, Chajia and Nfaoui [8] presented a thorough test of various classifiers on telecom churn data and stated that a Logistic Regression model attained higher accuracy than more intricate methods, such as KNN, SVM, and neural networks. Their results support the idea that simple models may outperform more complicated methods in cases when features are well designed, and data is well pre-prepared.

So far, the researches highlights an important trade-off, the more complex models (Deep neural network, Deep learning, XGBoost) may provide us with a slight accuracy improvement, but models such as Logistic Regression, Decision Tree, Random Forest, and even SVM were favored for their consistencies, interpretability, and trustworthiness in predictions. Notably, there are cases where simple model can outperform more complex ones. Chajia and Nfaoui [9] has experimented with various classifiers on telecom churn data by generating rich features using language model embeddings and found out that a plain logistic regression classifier actually got the best accuracy results at around 89%, outperforming KNN, SVM, and neural networks in their tests [8]. These comparative outcomes have demonstrated that while the advanced methods such as Deep Learning and boosted trees often lead to high accuracy, the classic models like Decision Trees, Logistic Regression, SVM, and Random Forests are not to be excluded and can even sometimes perform better especially when the data is prepared or features engineered effectively.

2.1.2 Handling Class Imbalance in Churn Data

Churn datasets usually suffer from class imbalances, which can cause model predictions to be skewed. Alif and Fahrudin [10] addresses this issue by applying oversampling and undersampling techniques such as SMOTE, ADASYN, and IHT to improve their model's specificity. Their analysis revealed that while oversampling can reduce overall accuracy, it also significantly enhances the model's ability to detect minority churn classes, particularly with the IHT method in Random Forest, this has shown to achieved a specificity of 93.4%. This emphasizes the need for sampling strategies in real-world churn scenarios. This class imbalance issue poses a persistent challenge for model training, as classifiers can become biased towards predicting the majority class [11]. Recent literature has emphasized that proper imbalance handling is just as important as the choice of algorithm. According to Sikri et al [11]

the “inherent challenge” of imbalanced churn data is highlighted and demonstrates that applying targeted data balancing significantly improves overall model accuracy [11].

According to Imani et al [12], they extended their analysis by comparing SMOTE and ADASYN on telco churn data across multiple advanced classifiers such as CatBoost and LightBGM. They also observed that while SMOTE has provided a more stable overall performance, ADASYN shown slight improvement in minority class recall, reinforcing the importance of choosing context-appropriate sampling techniques. Likewise, Pun et al [13] has demonstrated the effectiveness of ADASYN when combined with an ensemble of SVM, neural networks, and Logistic Regression, they achieved an AUC of 0.90.

The traditional classifiers we know can incorporate imbalance solutions to boost their effectiveness. Bhatnagar and Srivastava [5] used techniques such as SMOTE (Synthetic Minority Oversampling), edited nearest neighbour cleaning, and random undersampling in their workflow [5]. By combining these techniques with cross-validation and hyper-parameter tuning, they have ensured that their models like Logistic Regression, Decision Trees, SVM, and Random Forests were all trained on a more balanced representation of churn vs non churn customers [5]. The end result was a fair and comparison in which even simpler models could detect churn patterns rather than always defaulting to the majority class. Some modern algorithms like XGBoost or other boosting methods have inbuilt parameters to help handle class imbalance. However, the traditional models can handle skewed data equally with external balancing (either by cost-sensitive learning or resampling). The literature contends that data pre-processing is crucial and that no single model can resolve imbalance on its own. The minority class can achieve great recall even with simple classifiers like SVM or decision trees when imbalance is corrected. For instance, in Akbar & Apriono's experiment, the Decision Tree's AUC (~0.747) on churn prediction shows that, once the dataset was prepared and characteristics were normalised, it was catching churners almost as well as the more sophisticated XGBoost [7]. In short, recent telco churn studies have stressed that careful balancing and feature engineering will put logistic regression, decision tree, random forest, and SVM on a levelled playing field, this allows them to perform robustly despite the skewed class distributions.

Recent studies have shown to shift towards hybrid sampling strategies and model tuning to further improve churn prediction. According to Kimura [14] they have explored the use of SMOTE-Tomek Links and SMOTE-ENN in combination with ensemble models such as SVM

and Random Forest. The results they have obtained suggest that SMOTE-ENN is more effective in handling noisy data, while SMOTE-Tomek is better for datasets with overlapping classes. This difference is very important when selecting a resampling technique to be used based on the data structure.

Looking at the next point, Imani et al [15] have integrated hyperparameter optimization by using Optuna with hybrid sampling methods such as SMOTE, SMOTE-ENN, and SMOTE-Tomek. Their testing and experiments have demonstrated significant improvements in precision, recall, and F1-score, showing that by combining sampling strategies with optimized model configurations it results in a more reliable and accurate churn prediction.

Suguna et al [16] have also validated the performance of SMOTE when it is paired with ensemble learning techniques such as AdaBoost and Gradient Boosting. Their study found that by combining oversampling with ensemble methods, it enhances recall while maintaining competitive overall accuracy particularly beneficial in telco churn scenarios where minority detection is extremely important.

The findings above collectively supported the argument that proper data balancing significantly enhances model sensitivity to churners especially when its used together with robust classifiers

2.1.2.1 Oversampling Techniques

Oversampling is one of the most widely adopted strategies for addressing class imbalance, particularly in churn datasets where minority churners represent a small fraction of the population. Oversampling techniques increase the presence of the minority class in the training set by generating new synthetic instances or adaptively enriching sparse regions. The primary goal is to provide learning algorithms with a more balanced representation, enabling them to capture underlying churn patterns that would otherwise be overshadowed by the majority class. **SMOTE** remains the foundational algorithm in classical oversampling, generating synthetic minority samples through interpolation. While effective, SMOTE may introduce synthetic points in noisy or overlapping regions, which has led to the development of more advanced extensions.

ADASYN improves upon SMOTE by adaptively creating more synthetic samples for difficult-to-classify minority instances, particularly those near the decision boundary. This approach helps increase model sensitivity toward borderline churners.

K-Means SMOTE, a more modern technique, incorporates clustering prior to oversampling. By first grouping minority samples using k-means and then performing SMOTE within each cluster, K-Means SMOTE ensures that synthetic samples are generated in dense and meaningful regions of feature space, reducing noise and boosting recall and F1-score performance in churn and credit risk domains [17].

Recent developments integrate deep generative models into oversampling. **VAE-SMOTE** uses a Variational Autoencoder to learn the latent distribution of minority samples and generate new synthetic points that capture non-linear structures more effectively. Empirical studies show that VAE-based oversampling can outperform both GAN-SMOTE and classical oversampling techniques in recall and F1-score, especially in datasets with complex minority distributions [18].

Another promising generative method is **GAN-based oversampling**, such as the CTGAN-ENN hybrid model. CTGAN synthesizes realistic tabular samples, while ENN removes noise after generation. This combination has been shown to outperform SMOTE-ENN, ADASYN-ENN, and WGAN-GP on multiple churn datasets, with XGBoost and LightGBM models achieving F1-scores as high as 0.95, significantly surpassing many traditional oversampling approaches [19].

Collectively, these advanced oversampling techniques indicate a shift toward structure-aware and distribution-sensitive data generation, improving minority class detection and overall model robustness.

2.1.2.2 Undersampling and Hybrid Cleaning Techniques

Undersampling and hybrid cleaning methods address class imbalance by selectively removing redundant or noisy majority samples to reveal clearer decision boundaries. These approaches improve classifier performance by reducing the overwhelming influence of the majority class and sharpening the representation of minority churn instances.

Tomek Links identify majority–minority pairs that lie close together in feature space. Removing the majority instance from each Tomek pair helps eliminate overlapping class regions, improving the quality of the training set. **Edited Nearest Neighbours (ENN)** further cleans the dataset by removing instances misclassified by their nearest neighbours, often eliminating majority samples near noisy or ambiguous regions. These cleaning approaches are

commonly used after SMOTE to form hybrid methods such as SMOTE-Tomek and SMOTE-ENN.

NearMiss, one of the earliest undersampling techniques, selects majority instances that lie closest to minority samples, helping preserve the most informative majority examples. This is especially useful in churn prediction, where churners often lie near the boundary and require models to focus on subtle, discriminative decision regions. NearMiss has demonstrated improvements in minority class recall and G-Mean in various imbalanced datasets [20].

Recent research has also explored algorithmic approaches to evaluate data complexity.

Instance Hardness Threshold (IHT) removes majority samples that are too easy to classify, retaining only those likely to challenge the model. By focusing learning on difficult regions, IHT has been shown to enhance F1-score and minority recall in tabular imbalanced classification tasks, including churn prediction [21].

Although labelled as an oversampling method, **Majority Density Oversampling (MDO)**—particularly the HEOM-GA variant—operates as a hybrid approach that analyses the density of majority clusters and avoids oversampling in highly congested regions. Instead, it strategically removes or avoids generating points in majority-dense areas and produces synthetic samples only in sparse or borderline regions. Studies using HEOM-GA report significant improvements in recall, G-Mean, and AUC compared to SMOTE, ADASYN, and other oversampling baselines [21].

Together, these undersampling and hybrid techniques complement oversampling by cleaning and refining the decision boundary. When combined with powerful classifiers, they provide a more informative, balanced, and noise-reduced dataset, enabling superior churn prediction performance.

2.1.3 Interpretability and Explainability

The main reason for choosing Decision Trees, Random Forests, Logistic Regression, and SVM over the more complex models is the telco industry's requirement for interpretability. Business stakeholders need to understand why a model flags a client as likely to churn in order to take the proper retention measures. client churn management is not merely a technological exercise. This is often where traditional machine learning methods shine. The if-then rules from decision trees are clear to humans, and the results from logistic regression show how each attribute affects the likelihood of churn. Modern deep learning and ensemble techniques, on the other hand, frequently act as "black boxes," making it challenging for business analysts to believe or understand their forecasts. Imani et al underscore this issue in their article noting down that

even though advanced deep models can help improve accuracy, their **opaque decision processes** slows down their adoption in churn prediction systems from the article the authors have stated that “*neural networks and ensemble models function as black boxes, making it difficult for businesses to trust their predictions*” [22]. They also pointed out that despite the emergence of explainable AI techniques such as SHAP values and LIME, that these methods are “*not yet widely adopted in real-world churn prediction systems*” [22]. This gap properly highlights why many telco practitioners still stuck to the more interpretable models.

Multiple studies ranging from 2020 to 2024 has especially emphasized on the interpretability as an important deciding factor in model selections. Poudel et al. [6] has observed that previous churn prediction works often “*neglect the aspect of interpretability, vital for decision-making,*” and they tried to fix this by using explanation methods alongside their models [6]. In their experiments, although a GBM ensemble has achieved the best accuracy, the authors used SHAP plots and other visualisations to help telecom stakeholders understand what the model predicted [6]. This is part of a bigger trend that researchers feel like they need to add an explainability layer when they use a complicated model. Or one can choose a simpler model and get interpretability automatically. In their study on churn, Prabadevi et al. [8] say that Logistic Regression was the “most interpretable and reliable” model [9]. Its reliability most likely refers to the stability and clarity of its predictions; with a logistic model, we can easily see how features like tenure and contract may affect churn offs, and this transparency was valued even though a deep learning model had slightly higher raw accuracy [9]. Likewise, decision trees have always been popular especially in the telco churn projects for being able to produce clear decision rules. Imani [22] found out that by augmenting machine learning churn models with explainable rule-based components will improve user acceptance of the results [22]. The consistent message is that **interpretability matters**: organizations prefer models that not only predict well but can also articulate the reasons for churn – a strength of logistic regression and decision tree-based methods over techniques like ANN, CNN, or even highly tuned ensembles.

It is also worth mentioning that Random Forests and SVM, are among the “classic” models, are actually somewhat less interpretable than a single decision tree or logistic regression, yet they are still often favored by users over the complex deep learning in telco churn due to a balance of accuracy and partial explainability. Random Forest, being an ensemble of trees, does not result in simple rules, but it can also provide feature importance measures that is able

to highlight key churn drivers for example showing that contract types or tenure types are top predictors [7]. This level of insight is usually enough for operational use, even if the exact decision path for each prediction is not as straightforward as a single tree. SVMs, especially with non-linear kernels, are like black boxes in terms of direct interpretability, but during practice SVMs are usually applied with a limited number of features that are well understood, and domain experts are still able to reason about the support vectors or use techniques to visualize the decision boundaries. More importantly, both the Random Forests and SVMs have a long track record in churn prediction researches, because of this their behavior on churn problems itself is actually well studied and documented by many researchers, this lends a degree of familiarity and explainability by experience. In conclusion, recent literature have strongly advocated for interpretability in churn modeling. Traditional models such as Decision Tree, Logistic Regression, Random Forests, and SVM either offer inherent transparency or at the very least a middle ground of explainability, whereas newer classifiers like deep neural networks or even complex ensemble methods all require extra effort to interpret, which can be a technical barrier in the telco industry [22].

2.1.4 Operational and Business Considerations

Beyond the performance metrics and interpretability, practical deployment factors play a crucial role in model selection for a telco churn project. Telco companies often deal with large scale data and needs solutions that will integrate with existing systems such as CRM databases and real time dashboards and operate within its own computational constraints. In many cases, many companies resort to a simpler model because they are more operationally feasible, they are easier to implement, faster to train, and also much less demanding in production. Imani et al [22] has pointed out that many studies that have been done “*focus on improving model accuracy*” but they seem to always give lesser attention to the “*real world deployment challenges*”, such as a company’s ability to scale and integrate [22]. They also noted that “*many organizations lack the computational infrastructure to support complex DL models*”, and emphasized the need for a churn prediction solution that is lightweight, efficient, and real time deployable [22]. This observation has directly justified why a company will deliberately choose simpler models like logistic regression or decision trees over more complex and cutting-edge methods like deep neural network, because the latter might simply be impractical to deploy in live telco environment that requires quick and precise predictions or has limited GPU resources.

Operational relevance also includes the ease of updating and maintaining the model. Logistic Regression and Decision Tree models are relatively straightforward to retrain periodically with new data, and they also often require fewer hyperparameter tuning efforts compared to boosting algorithms or deep learning models. This is because both the models mentioned are less sensitive to parameter initialization and can be retrained quickly, which is extremely important as telco customer behavior evolves over time depending on various factors like new competitors, services, etc. Even though SVMs and Random Forests do involve some parameter tuning (e.g. kernel choices for SVM, number of trees for RF), but these models still have implementations in most data science frameworks that are actually well understood, and they can also be scaled to moderately large datasets. For example, Random Forest has been successfully applied to telco churn datasets with millions of records by leveraging its parallel tree training, whereas a comparable neural network might require much more detailed optimization and distributed computing to reach the same scale [22]. In a study by Ullah et al. [2], they observed that Random Forest achieved the best accuracy but it took longer to train as compared to a simple tree, yet it was still far more efficient than training a multi-layer neural network on the same data [2]. More recent sources have found out that computational efficiency favors the classical machine learning for many of the churn tasks. For instance, a study from 2023 has reported while XGBoost was accurate, it was actually outpaced in execution speed by a simpler Decision Tree and Naive Bayes model which processed the churn data much more quicker while yielding the same AUC scores [7]. When it comes to real life business operations like the telco industry, these differences matter: a slightly lower accuracy model that can be deployed immediately and refreshed frequently may be a more preferable approach to a marginally more accurate model that is complex and expensive to maintain.

Another important operational factor is the alignment with business objectives and with the ability to incorporate business constraints. The logistic regression model is inherently a probability-based model, which makes it more convenient to be implemented with cost based thresholds. For example, a company can use logistic output probabilities to decide whether if a customer's churn risk exceeds a certain monetary threshold in order to trigger a retention offer. This kind of cost-sensitive decision role is actually easier to derive from a probabilistic model like the logistic regression model. While any model's output can be converted into probability, the logistic model is naturally more interpretable in those terms and its calibration is also more straightforward, which could be one of the reasons some telco churn studies stick with logistic regression for operational decision making [8]. Models like Decision Trees on the

other hand can directly encode business rules and constraints in their structure which is something that companies might find useful if they want or need a model that can be translated into a set of if-then rules for their call center staff or an automated retention system. These pragmatic concerns of **speed, scalability, maintainability, and alignment with business processes** often tilt the balance in favor of classical ML models over more complex ones in churn prediction [22].

2.1.5 Summary

This literature review demonstrates that churn prediction has evolved from basic decision trees and logistic regression models to more advanced, hybrid, and ensemble-based methods. Random Forest still seems to be the top performer in terms of classification accuracy, while usage-based models and data balancing techniques enhances the interpretability and minority class detection. However, there exists gaps in applying these models to real-time systems and in achieving a balance between accuracy and explainability.

Not only that these findings ranging from 2020 – 2025 have also shown strong justification for using simpler models such as **Decision Trees, Logistic Regression, Random Forests**, and **SVMs** as the core models in the telco customer churn prediction project. While other complex and advanced techniques like XGBoost, artificial neural networks (ANNs), and deep learning can sometimes deliver a much higher accuracy result [6], [8], the improvement often comes at the cost of interpretability, computational complexity, cost, and deployment hurdles. The chosen classic models each offer a compelling advantage, the table below will simplify their advantages according to the articles reviewed:

Model	Key Strengths
Decision Tree	Transparent rules; easy to interpret by business users
Logistic Regression	Statistically precise; interpretable probability estimates for churn risk
Random Forest	High accuracy and robustness; able to handle complex interactions; resists overfitting
Support Vector Machine (SVM)	Handles non-linear patterns; efficient with feature engineering

Table 1. Summary of Justified Model Strengths for Telco Churn Prediction

Table 1 summarized the key strengths of the selected models used in this telco churn prediction project. Decision tree models are highlighted for their transparent rule-based structure, which allows business users to easily interpret churn decisions. Logistic Regression is used here for its statistical precision and ability to produce interpretable probability estimates, making it very suitable for risk assessments and decision support. Random Forest is noted for its strong predictive accuracy and robustness, particularly in handling complex feature interactions while reducing overfitting through ensemble learning. Support Vector Machine (SVM) is included for its effectiveness in modeling non-linear decision boundaries and its efficiency when combined with appropriate feature engineering. Collectively, the models listed in Table 1 provide a balanced trade-off between predictive performance, interpretability, and practical deployment suitability.

Not only that, researchers have shown that with proper attention to class imbalance by means of re-sampling or cost-sensitive learning, these traditional models can still achieve higher recall and precision for the minority class [5], [11]. They strike a balance between performance and practicality. As Imani et al. [22] conclude in their survey, the future of churn prediction lies in “*developing robust, interpretable, economically beneficial churn prediction models*” that aligns with business goals [22]. In the telco industry, the usage of interpretable machine learning models (augmented if necessary, with explainable AI techniques) is aligned with this vision, providing models that not like predict churn with competitive accuracy, but it will also be able to generate insights and actions that telco operators can trust and act upon. The evidence gathered thus far have suggested that the four highlighted models is able to fulfill these criteria, which is why many recent studies and practical deployments continue to rely on them in favor of the more complex models [8], [22].

So this project aims to build upon these findings by evaluating several classification models, incorporating sampling techniques, and identifying the most significant features that contribute to churn.

3.0 Methodology

This section details the step-by-step approach adopted to design, build, and evaluate machine learning models for customer churn prediction in the telco sector. It also covers the data collection strategy, preprocessing workflow, feature engineering, algorithm implementation, and the evaluation framework used to measure model effectiveness.

3.1 CRISP-DM

CRISP-DM stands for cross-industry process for data mining. The CRISP-DM methodology provides a structured approach to planning a data mining project. It is a robust and well-proven methodology [23]. CRISP-DM is widely recognized for guiding analytical projects through six distinct phases: business understanding, data understanding, data preparation, modelling, evaluation, and deployment. Each phase is tailored to address the specific objectives of this telco churn analysis project.

The table below outlines how each CRISP-DM phase will be implemented in this project:

CRISP-DM Phase	Description	Application in This Project
Business Understanding	Define the business goal and project objectives.	The main goal is to predict telecom customer churn to help the company proactively manage customer retention.
Data Understanding	Collect and explore data to identify structure, quality issues, and initial insights.	Loaded the Telco Churn dataset. Conducted exploratory data analysis to understand churn distribution and patterns.
Data Preparation	Clean, transform, and prepare the dataset for model training.	Categorical features were encoded, missing values handled, numerical features scaled, and class imbalance addressed using SMOTE and IHT.
Modelling	Apply and train suitable machine learning models.	Four algorithms were applied: Decision Tree, Random Forest, Logistic Regression, and SVM. Hyperparameters were tuned using cross-validation.
Evaluation	Assess the performance of	Models were compared

	models using evaluation metrics.	using accuracy, precision, recall, F1-score, and confusion matrix. Random Forest gave the best result.
--	----------------------------------	--

Table 2. CRISP-DM Cycle

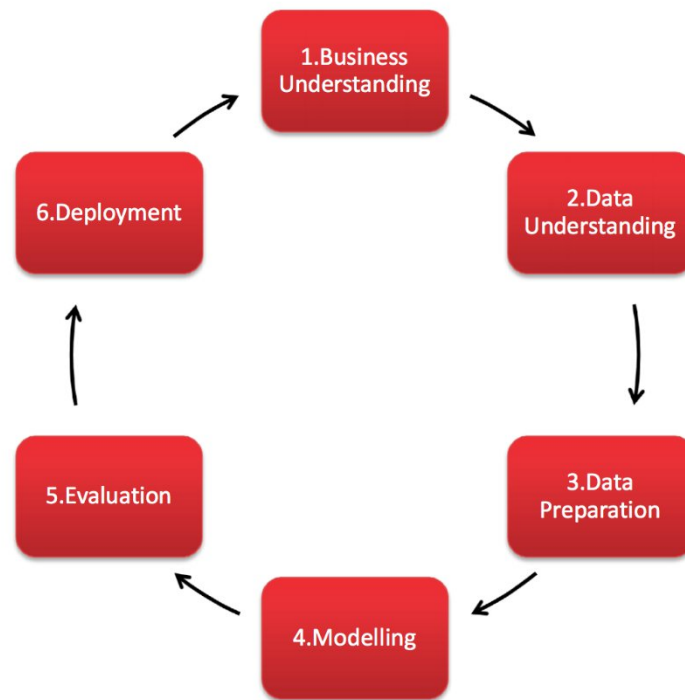


Fig 1. Image of CRISP DM Phases [23]

3.2 Business Understanding

Telecommunication companies face intense competition, and customer churn leads to significant revenue loss. This project aims to develop a predictive model that can identify customers likely to churn, enabling the business to implement proactive retention strategies.

Business Goals:

- Reduce churn rate by identifying at-risk customers early
- Improve decision-making in retention campaigns
- Prioritize high-value customers in churn intervention
- Provide explainable outputs (e.g. feature importance) to support business decisions

Project Success Criteria:

- High recall and F1-score for the **churn (minority) class**
- Interpretability of model decisions for operational integration
- Model performance evaluated on real-world metrics like AUC, precision, and recall

3.3 Data Understanding

The dataset that will be used for this project is sourced from [Kaggle](#). It includes over 7,000 customer records, each containing demographic, behavioural, and account-related information. Features that are included in the dataset are such as gender, tenure, monthly charges, contract type, payment method, internet service, and whether the customer has churned. The target variable is binary, indicating if the customer churned ("Yes") or not ("No") as sort of a Boolean.

A critical characteristic of this dataset is its class imbalance; this is caused by the 26% of the churned customers that are recorded which places it in the minority class while the remaining 74% of customers did not churn which places it in the majority class. This 1:3 imbalance ratio can definitely bias models towards always predicting the majority class. Therefore, this dataset requires steps to perform oversampling and undersampling to further improve its predictions on minority classes. Due to this, addressing class imbalance is a key component of the methodology.

In preparation of the dataset a few important things must be done, of which are data encoding. It refers to the process of converting categorical or textual data into numerical format, so that it can be used as input for algorithms to process [22]. This allows most of the machine algorithms used to work with numbers and not just texts. Then data splitting which is a crucial step to train data. The dataset was partitioned into training (80%) and testing (20%) subsets to enable model validation. The motivation for using an 80 / 20 split is loosely driven by the Pareto principle (also called the 80–20 rule), which states that 80% of the effect is driven by 20% of causes (and vice versa). The Pareto principle isn't a mathematically guaranteed property, but many observed phenomena follow the Pareto principle [23].

Below are the variables along with its descriptions:

Dataset features:

- i. customerID – Unique identifier assigned to each customer.
- ii. gender – The customer's gender (Male or Female).
- iii. SeniorCitizen – Indicates if the customer is a senior (1 = Yes, 0 = No).
- iv. Partner – Whether the customer has a partner (Yes/No).
- v. Dependents – Whether the customer has dependents (Yes/No).
- vi. tenure – Number of months the customer has stayed with the company.

- vii. PhoneService – Whether the customer has phone service.
- viii. MultipleLines – Whether the customer has multiple lines.
- ix. InternetService – Type of internet service subscribed to (DSL, Fiber optic, None).
- x. OnlineSecurity – Whether the customer has online security add-on.
- xi. OnlineBackup – Whether the customer has online backup service.
- xii. DeviceProtection – Whether the customer has device protection service.
- xiii. TechSupport – Whether the customer has tech support.
- xiv. StreamingTV – Whether the customer streams TV services.
- xv. StreamingMovies – Whether the customer streams movies.
- xvi. Contract – Type of contract (Month-to-month, One year, Two year).
- xvii. PaperlessBilling – Whether the customer receives paperless billing.
- xviii. PaymentMethod – Customer’s payment method (e.g., bank transfer, credit card).
- xix. MonthlyCharges – Amount charged to the customer monthly.
- xx. TotalCharges – Total amount charged to the customer to date.
- xxi. Churn – Target variable indicating whether the customer has churned (Yes/No).

This dataset consists of customer demographic details, service subscriptions, billing information, and churn status, it offers a comprehensive description of customer behavior in a telco company. It also enables classification modeling to predict churn probability and is widely used for customer retention analytics. In this project, it will be used to develop, evaluate, and compare machine learning models under different sampling strategies to address class imbalance.

3.4 Data Preparation

We now can outline the overall procedures of how these components fit together in our methodology starting with **data splitting**. The dataset is first split into training and test sets by following the widely adopted 80/20 rule inspired by the pareto principle [23], where 80% of the data is used to train the model and 20% is reserved for unbiased evaluation. Stratified sampling is used to preserve class proportions. Alternatively, k-fold cross-validation may be applied on the training set for more robust performance estimates. It is important that oversampling is confined to the training data set only in order to prevent data leakage, no synthetic data is added to the test set and the test set remains imbalance as in reality.

Then the next step will be **applying oversampling**: On the training data, we will apply each of the oversampling methods in turn to create multiple balanced dataset for training. One training set will be oversampled using **SMOTE** to balance out the class distribution accurately by increasing its minority count to match the majority count. The other training set will be processed by **SMOTE-Tomek**, meaning the training set that is oversampled will be followed by Tomek link removal of the overlapping majority samples. Another training set is going to be processed by **SMOTE-ENN**, which is SMOTE oversampling followed by ENN to clean the misclassified instances as discussed above. The last oversampling method will be used is **ADASYN**, which adaptively generates synthetic samples for the minority class. By doing this we ensure the oversampling is done only on the training portion every time. The results is that the four different training sets are balanced in their own way but also has the potential to have different synthetic data and cleaned majority data.

Next will be **model training**. Each of the four classifiers which are Decision Tree, Logistic Regression, Random Forest, and SVM are trained on each of the balanced training dataset. During training, the standard hyperparameter settings are used initially (e.g., default tree depth, default SVM C and kernel, etc.), but a grid search and optimization may be performed if needed to ensure each model is reasonably tuned. Most importantly, the models have to see a 50:50 balance class distribution in their training data due to the oversampling. As for the hybrid methods, the training data size might be slightly reduced by means of SMOTE-Tomek, SMOTE-ENN removing some examples, but it is still balanced. We expect that, for example, the decision tree will no longer create biased splits since the classes are balanced [24]; the logistic regression will adjust its coefficients in order to separate the classes rather than default to majority class; the random forest model is expected to have trees that each get a balanced bootstrap sample; and the SVM will find a hyperplane that maximizes margin without being skewed by the class imbalance issue.

Lastly the **Comparison and Analysis** step. In this final step, we will be analyse which oversampling method results in the best improvement for each models. Some patterns are expected: for example, the more aggressive cleaning method of SMOTE-ENN might help the models that are sensitive to noise (perhaps benefitting SVM or logistic regression), whereas the simpler sampling method SMOTE might just be sufficient for robust models like random forest. ADASYN's focused approach may specifically improve recall if there are a few very hard minority examples that needs extra emphasis. Then, the baseline of models that are trained

on the original imbalanced data will also be compared to quantify the gain from these methods. This usually involves training each model on the unmodified training set and evaluating with the same metrics, likely those results will show lower recall and F1 for the minority class. By comparing them, we can display how much each resampling technique has helped to increase in minority class recall from say, about 50% to 90% for SMOTE for decision tree.

3.5 Model Development and Resampling Techniques for imbalance

I have researched and investigated four sampling strategies: SMOTE, SMOTE + Tomek Links, SMOTE + ENN, and ADASYN. All four of these methods tackle the class imbalance issue by creating a synthetic minority sample, sometimes combined with data cleaning to improve the overall quality of the training set.

3.5.1 Synthetic Minority Over-sampling Technique (SMOTE)

SMOTE is a seminal oversampling method that generates a synthetic minority class example to balance out the dataset [25]. Unlike random oversampling technique which duplicates minority instances, SMOTE creates newer instances by interpolation. When given a minority sample, SMOTE attempts to find its k nearest minority neighbours in feature space and then randomly selects one of these neighbours. A new synthetic instance is created along the line segment connecting the original sample and the neighbour [21]. In practice, usually one would take the feature vector difference between a minority example and one of its neighbours, multiplies it by a random number between 1 and 0, and adds it into the minority example to form a new synthetic point [21].

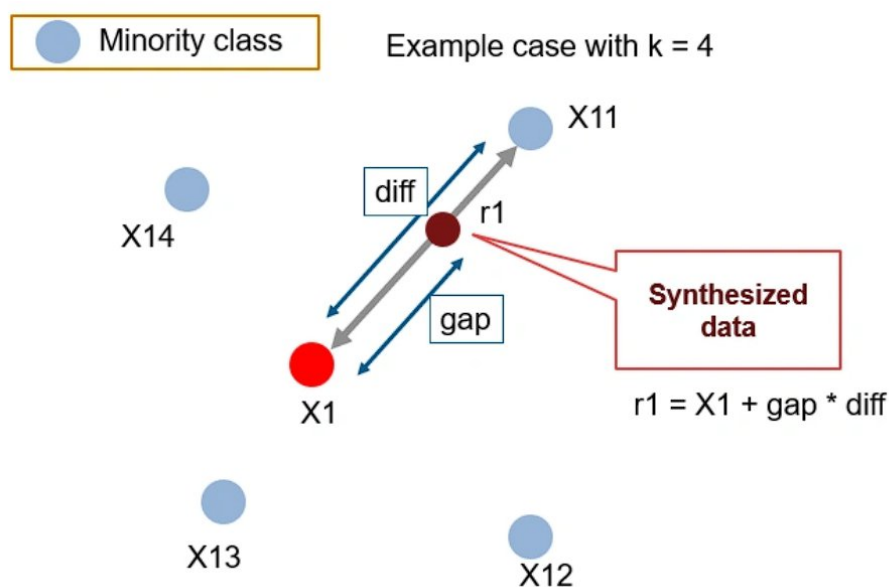


Fig 2. Illustration of SMOTE Sample Generation Process [25]

As we can see the image above, it illustrates how the SMOTE algorithm produces a synthetic sample for the minority class to tackle imbalance issues. In the example shown, a minority class instance which is shown as **X1** alongside its nearest neighbors which are labeled as **X11**, **X12**, **X13**, **X14** with $k = 4$. SMOTE selects a random neighbor and creates a synthetic data point **r1** along the line segment that connects **X1** and **X11**. This is done by computing the vector difference between the two points and multiplying it by a randomly generated scalar as seen in the figure. This technique has shown to be effective in diversifying the minority class by interpolating new samples within the feature space, rather than replicating existing data, which helps prevent overfitting and improves the model's ability to generalize [25].

This procedure effectively introduces new minority samples that are similar to the existing ones, thus it increases the diversity of the minority class data without simply replicating points. By spreading out the minority examples in feature space, SMOTE can help reduce the risk of overfitting that usually occurs with naïve oversampling [21]. SMOTE has been widely used and extended in many studies due to its balance of simplicity and effectiveness [21][22].

However, SMOTE creates synthetic data based on existing samples. If the original data is noisy or complex, it can introduce noise into the model, which may hurt performance [23]. SMOTE could generate synthetic points that are too similar to existing data, leading to overfitting [23] which might shift decision boundaries sub optimally [23]. Due to this, many researchers have proposed combining SMOTE with data cleaning techniques to remove borderline or noisy instances after oversampling [25]. Two such hybrid approaches will be discussed next which are SMOTE with Tomek Links and SMOTE with Edited Nearest Neighbours (ENN).

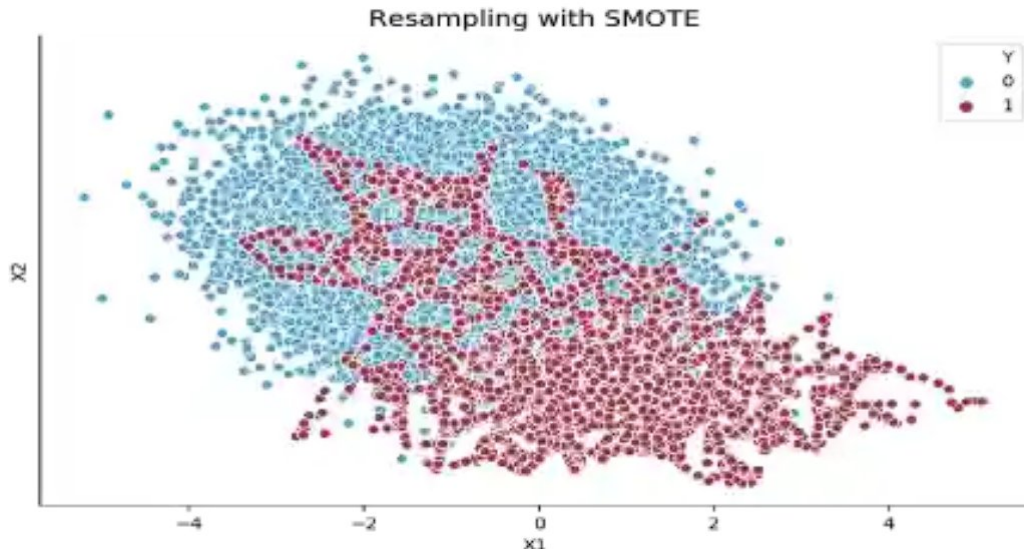


Fig 3. Resampling with SMOTE [25]

3.5.2 SMOTE with Tomek Links (SMOTE–Tomek)

SMOTE-Tomek is known as a hybrid resampling technique that derives from SMOTE itself, this technique applies SMOTE oversampling followed by Tomek link undersampling which is a removal technique to clean the dataset [23]. A Tomek link is a pair of one majority class instance and one minority class instance that are each other's nearest neighbours in a feature space [26]. These pairs typically indicate overlapping or obscure regions between classes, if two samples from opposite classes turned out to be very close, it is often harder for a classifier to distinguish them.

According to Tomek [26] the idea that one can remove these obscure pairs to clean the decision boundary is viable. In practice, the Tomek link algorithm will identify all such nearest neighbour pairs of opposite classes and remove one or both of the instances in that pair in order to eliminate overlap [26]. Usually, one would remove the majority class instance of each Tomek pair, as it is assumed to be an over overlapping majority example on the decision boundary. By doing this, we eliminate noisy boundary majority points and create a clearer separation between classes [26]. In SMOTE-Tomek, after generating synthetic minority samples via SMOTE, the combined dataset is then processed to remove the Tomek links which results in majority instances in Tomek pairs to be dropped [26]. This results in a more refined training set, in which the minority class has been amplified, and the obvious overlapping samples have been filtered out. Because of this SMOTE-Tomek addresses both quantity and

quality of training data which is done through oversampling and data cleaning respectively. In some effect, SMOTE-Tomek undersamples the majority class in a targeted way by removing the borderline majority examples that could mislead the classifier. When the minority class is very sparse and surrounded by majority points, this method especially can be used to improve classifier performance, as this method clears a buffer around the minority examples while it also adds new synthetic minority points.

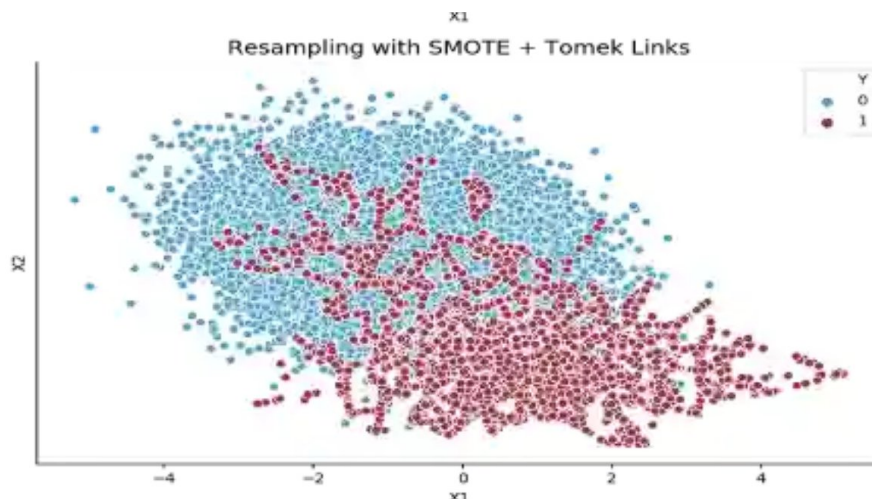


Fig 4. Resampling with SMOTE-Tomek [26]

3.5.3 SMOTE + Edited Nearest Neighbours (SMOTE-ENN)

Another hybrid approach that we can use is called SMOTE-ENN which combines SMOTE oversampling with the Edited Nearest Neighbours (ENN) data cleaning method. ENN, originally due to Wilson [27], is an algorithm that removes any instances whose class label is misclassified by its nearest neighbours. So in the context of imbalanced data, ENN is often used to trim away majority class instances that invade minority class regions and even in some cases, noisy minority instances by looking at local neighbour relationships. The SMOTE-ENN process works as follows: it first applies SMOTE to create synthetic minority examples as described earlier. Then, applying the ENN rule to the modified dataset. For each example in the dataset, look at its k nearest neighbours which is commonly known to be $k=3$. If an instance's class label differs from the majority class of its three nearest neighbours, that instance is then considered as misclassified by its neighbourhood and is removed.

In implementation, this typically will lead to the removal of majority class samples that are surrounded by minority neighbours since those majority samples would be misclassified by neighbour voting. Not only that, it can also remove a minority instance that is in a heavily majority neighbourhood, but some variations keeps all the minority points and only removing

conflicting majority points. Overall, the ENN method is more aggressive than Tomek links in cleaning data because it tends to remove more instances as it does not only consider pairwise neighbours but also the local neighbour consistency. By combining a string oversampling method with strict cleaning methods, SMOTE-ENN produces a dataset where the minority examples are synthetically increased, and many majority examples that are near the class boundary are eliminated. This often results in an even cleaner class separation than SMOTE-Tomek, at the cost of maybe removing a larger portion of the majority class thus reducing the training set size significantly [27].

The result is a training set that is balanced in quantity and also has lesser class overlap or noise. Prior studies have found that SMOTE-ENN can improve classifier performance on highly imbalanced problems, especially where the minority class has very few examples and the majority class contains borderline or mislabelled instances [20]. These combined methods underscore the importance of noise-filtering after oversampling: since SMOTE would blindly introduce synthetic points, applying ENN or Tomek helps to ensure that these new points are not right next to majority samples that would have confused the classifier.

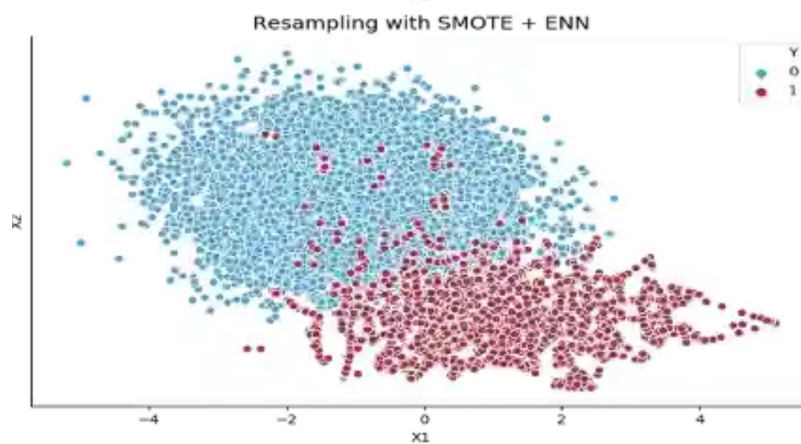


Fig 5. Resampling with SMOTE-ENN [27]

3.5.4 Adaptive Synthetic Oversampling (ADASYN)

While SMOTE and its combinations of the mentioned data cleaning methods is able to generate synthetic samples uniformly for all minority examples, ADASYN which is also known as Adaptive Synthetic Sampling, is a more recent technique that **adapts** the oversampling rate to the local difficulty of each minority example. He *et al.* [28] proposed that ADASYN builds on the concept of SMOTE but also introduces a density-based weighting: minority instances that are much harder to learn, for example like those in sparse regions or near to class boundaries

will receive more synthetic samples than those that are well surrounded by other minority points. The algorithm computes for each minority instances a ratio of how many of its k nearest neighbours belong to the other class; instances with higher proportion of majority neighbours are considered harder because they are more at risk of being misclassified and are also weighted to generate more synthetic examples. This will result in a more focused oversampling: regions of minority space that are easy which refers to high minority density will not be oversampled as much as the challenging regions which refers to high minority density or high overlap.

By doing so, ADASYN aims to reduce the learning bias introduced by the class imbalance issue in a data-distribution-sensitive way [28]. It has the effect of not only balancing the class counts but also able to reduce the bias and **adaptively** improve the classifier's attention to hard cases. One consequence of using ADASYN is that it may generate fewer total synthetic examples than SMOTE for the same imbalance ratio if many minority points are in dense clusters since it does not oversample those as much [28]. The creators of ADASYN reported improved performance across various metrics such as precision, recall, AUC by using this adaptive strategy. In short, ADASYN automatically prioritizes minority examples that would otherwise be overshadowed by the majority, thereby hopefully improving minority class recall without adding noise.

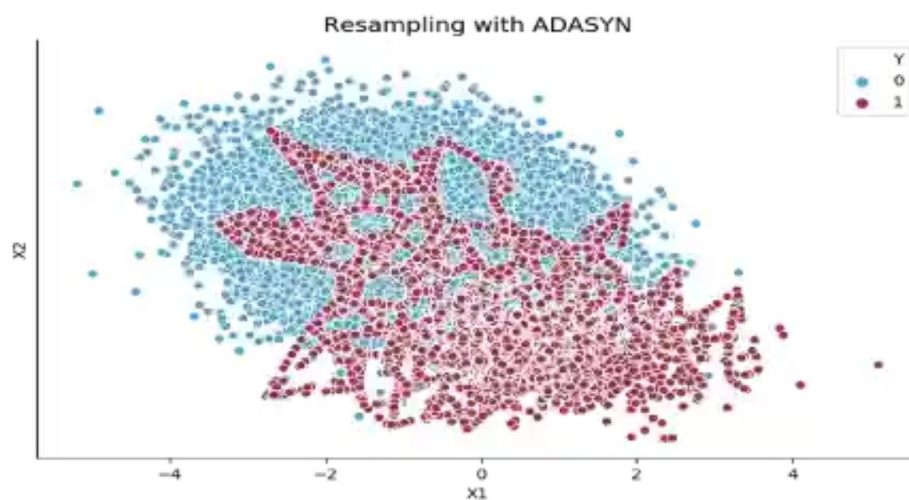


Fig 6. Resampling with ADASYN [20]

Each of the sampling methods discussed above will be applied in order to create a more balanced training dataset to be used in the churn analysis. Notably, methods like SMOTE-Tomek and SMOTE-ENN integrate an element of undersampling (cleaning) as well, this combination approach is powerful for dealing with cases where the minority class is not only

small but also surrounded by majority class noise [20]. Past research has shown that such hybrid methods can often yield better defined class clusters and improved model accuracy on imbalanced data [20]. Then a comparison will be made to compare the effectiveness of plain SMOTE, SMOTE-Tomek, SMOTE-ENN, and ADASYN in improving classifier outcomes.

Besides the main resampling methods listed above, some auxiliary sampling procedures were also added to be compared. They are Cluster-SMOTE, which uses k-means clustering before synthetic oversampling to maintain local data structure and Instance Hardness Threshold (IHT), which eliminates majority-class samples that are simple to make predictions on to target more challenging decision areas. In addition, NearMiss (v1) was used as a representative undersampling method to investigate the impacts of majority-class aggressive reduction on the model performance. The inclusion of these methods was a move to offer a more broad view on the way various sampling philosophies can affect the results of churn prediction.

3.6 Classification Models

After applying the aforementioned sampling techniques to address the imbalance issue that are present in the dataset, we can train and evaluate several classification algorithms. The models that are chosen for this project are Decision Tree, Logistic Regression, Random Forest, and Support Vector Machine (SVM). In this section, the topics that will be covered will be each model's theoretical basis and characteristics, as relevant to the methodology.

3.6.1 Decision Tree Classifier

A Decision Tree is a non-parametric, hierarchical model that recursively partitions the feature space to classify instances [29]. It is essentially a flowchart-like tree structure where each internal node represents a decision on a feature which is a splitting rule, each branch corresponds to an outcome of the decision made, and each leaf node represents a class label or a distribution over classes. The tree is constructed by choosing splits that maximize the homogeneity of the target class that is within the resulting subsets which is a common criteria for splitting include *Gini impurity* or *information gain (entropy)*. The process, which is also known as a recursive partitioning, continues until it further splits, provide little value, or a stopping criterion is met [29].

Decision trees are complimented for their intelligibility and simplicity: the model's decision rules can be easily understood and interpreted by humans and visualised [29]. Decision trees can also handle both numerical and categorical data and require minimal preprocessing which

means there is no need for feature scaling and can handle missing values in some implementations. One important aspect of decision trees is their tendency to overfit if it is grown too deep for example it will create overly complex rules that memorises the training data. This is usually prevented or mitigated by pruning or by setting certain constraints like maximum depth or minimum samples per leaf.

However, relevant to the project, decision trees can be biased by imbalance if used on a unbalanced dataset. If one of the classes dominates the other, a tree will tend to create branches that favours predicting that class, because the impurity measures are massively skewed by the class frequencies. For example, a tree might make a decision that yields a large pure majority class node while ignoring a small minority segment, since reducing impurity for the large segment has more impact on the objective. It has been noted and recorded that “decision tree learners create biased trees if some classes dominate”, and it is recommended by many to balance out the dataset prior to training a decision tree [24].

It is true that many tree implementations allow setting class weights or use criteria that adjust for class frequency, but in the methodology, it is already addressed by applying the oversampling techniques earlier. By supplying a balanced training set via SMOTE, etc, we ensure that the decision tree has an equal opportunity to grow branches for minority class patterns. A balanced dataset helps the tree to better identify discriminative splits for the minority class, rather than treating those instances as noise. In conclusion, the decision tree provides an interpretable baseline model that can capture non-linear relationships, and using oversampling is expected to significantly improve its minority class performance [24].

3.6.2 Logistic Regression

Despite its name, Logistic Regression is actually a linear classification model for binary outcomes. In the context of telco churn prediction, it estimates the probability of a customer belonging to the minority churn class by applying a logistic (sigmoid) function to a linear combination of input features [30]. The sigmoid function maps real-valued inputs into a probability range between 0 and 1, enabling probabilistic interpretation of model outputs [31].

$$Y = P(x) = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}}$$

Fig 7. Logistic Regression probability equation[32]

In other words, logistic regression, the model computed a linear combination of inputs as $\mathbf{w} \cdot \mathbf{x} + \mathbf{b}$, which is then passed through the sigmoid function $\sigma(t) = 1/(1+e^{-t})$ to obtain a probability score between 0 and 1. A customer is classified as a churner when the predicted probability exceeds a predefined threshold, typically set at 0.5 [33]. The model's parameters are typically estimated by maximum likelihood, equivalently minimizing the logistic loss, also known as cross-entropy on the training data. Despite its relative simplicity, logistic regression is a sound statistical model because it makes the assumption that the two classes are separated by a linear decision boundary in feature space, or a hyperplane. Efficiency, the ability to output calibrated probabilities, and the lack of need to adjust numerous hyperparameters (possibly except for regularization strength) are some of its advantages.

In the context of imbalanced data, logistic regression can suffer because it emphasizes overall accuracy unless instructed otherwise. If the classes are very imbalanced, a plain logistic model might resort to a solution that essentially ignores the minority class by predicting the majority for most of the input space since that results in minimal errors in terms of absolute number of mistakes. One solution to this is by incorporating class weights or using oversampled data so that the model sees more minority examples during training. In the methodology, by training logistic regression on the resampled datasets which are balanced by SMOTE or its other hybrid variants, we can provide the model with an equal number of positive and negative instances, which steers the optimization to find a decision boundary that is more effective in capturing the minority class.

Essentially, oversampling will prevent the scenario where the best fit is to always predict the majority class due to its nature. Since logistic regression is a linear model, its decision boundary's capacity is limited, it works best if the classes may not capture complex non-linear patterns that a decision tree or random forest can; however, it can be served as a helpful benchmark and is less prone to overfitting. We use logistic regression as a baseline parametric classifier, and we anticipate that the sampling techniques (as well as suitable assessment measures that concentrate on minority predictions) will largely increase its performance on unbalanced data.

3.6.3 Random Forest

A Random Forest is an ensemble learning method that operates by constructing a large number of decision trees and combining their results. It was introduced by Breiman [31] as an improvement over just simple bagging of decision trees. In a random forest classifier, each tree is trained on a bootstrap sample which is a random sample with replacement of the training data, and additionally, each split in a tree is determined by considering a random subset of the features rather than all of its features [34]. This randomization in both data and feature selection results in an ensemble of decorrelated trees. At prediction time, the forest takes a majority vote among all of its trees for classification to decide the final class [34]. By averaging many deep decision trees, the random forest achieves a lower variance model while maintaining low bias [34]. In a sense, it reduces the tendency for decision trees to overfit while a single tree may overfit its training set, the average of the many other trees (each trained on different data) smooths out the noise and results in a more robust generalization [34]. Random forests have become more and more popular for their strong predictive performance out-of-the-box and their ability to handle high-dimensional data and complex feature interactions.

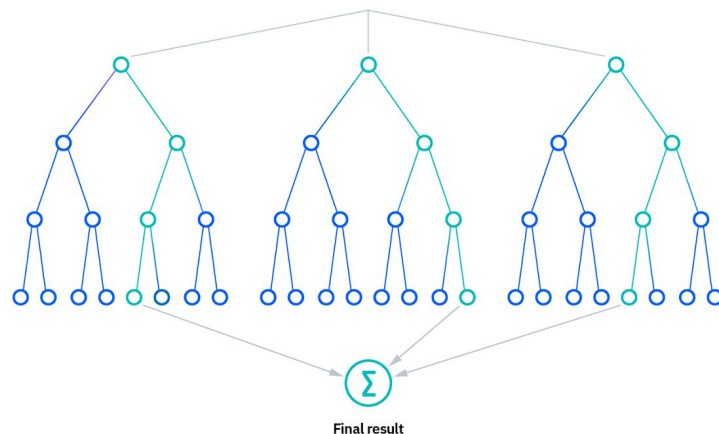


Fig 8. Representation of Random Forest [35]

For this project, random forest represents a more powerful classifier that is able to capture non-linear decision boundaries and interactions, which may be beneficial for differentiating minority class instances that a linear model could not. Random forests are somewhat invariant to scaling and can also naturally handle unbalanced data better than a single tree because of the ensemble vote can correct some bias; however, they can still be biased if the dataset is extremely imbalanced. By default, each of the trees in the forest is grown to minimize impurity without regard to the class frequency, so like decision trees, a forest can still favour towards the majority class in order to reduce overall impurity. Impurity is a combination of the existing

label on the node; the more heterogenous the split, the more impure the node was. In the case of Random Forest, the feature MDI is calculated as the average of the impurity decrease for that feature over all the trees in the forest [35].

In our methodology, we have applied oversampling prior to training the random forest, so that each tree of the forest is trained on a balanced bootstrap sample since the training set given to the algorithm is already balanced. This way, the random forest receives plenty of minority class examples in its bootstraps, which should improve its ability to identify minority patterns. Past research indicates that ensembles like random forests often perform well on imbalanced sets when combined with oversampling [20], in part because the oversampling can be seen as a way of informing each tree that the minority instances are as important as the majority ones. Another advantage is that because random forests average, they are less likely to overfit on replicated minority cases; each synthetic example from SMOTE or ADASYN will only impact a subset of the trees. All things considered, we expect the random forest to yield outstanding outcomes, and it will be intriguing to see how much the sophisticated sampling strategies improve its performance.

3.6.4 Support Vector Machine (SVM)

A Support Vector Machine is a powerful classifier that searches for the optimal hyperplane to separate classes by maximizing the margin between them. In the linear separable case, SVM will choose the hyperplane that leaves the largest margin from the nearest training points of any class [36]. Those nearest training points that lie on the margin are known as the support vectors, they essentially “support” or define the position of the separating hyperplane [36].

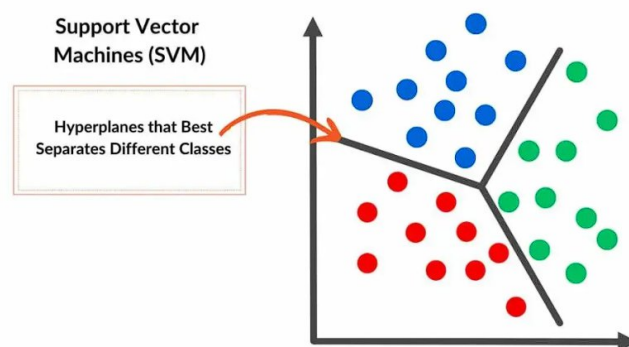


Fig 9. Representation of Support Vector Machines [37]

By maximizing the margin, SVM aims to achieve good generalization, as a larger margin usually correlates with lower classifier complexity according to the VC dimension theory [36]. The standard SVM is a binary linear classifier, but through the use of kernel functions it can efficiently handle non-linear classifier tasks [36]. By using kernel functions to compute dot products in the implicit feature space, SVM is able to project the input features into a higher-dimensional space (possibly infinite-dimensional) where a linear separator may exist without explicitly computing the coordinates in that space [36].

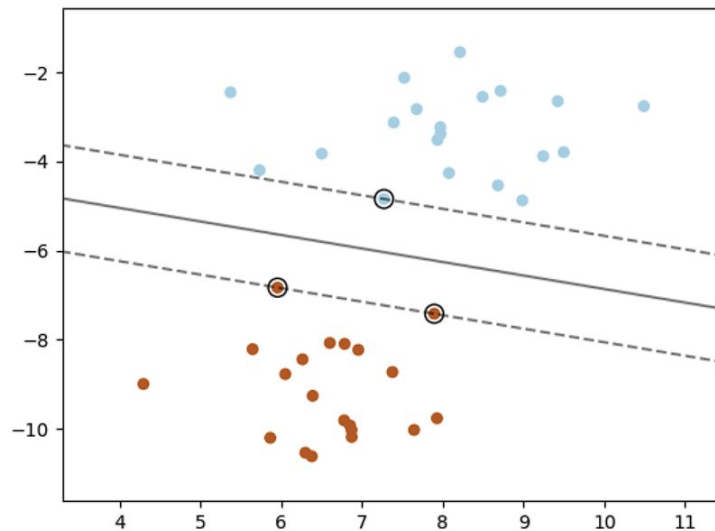


Fig 10. SVM classifier with support vectors and maximum-margin hyperplane [38]

An example of a Support Vector Machine classifier identifying the hyperplane with the largest margin. Two potential separating lines in this 2D example are H1 and H2, but H2 obtains a greater gap between the two classes (distance to nearest points is maximal). The support vectors that define the ideal hyperplane are represented by the points on the margin (circled).

When the parameter C is properly adjusted, SVMs are known to be efficient in high-dimensional spaces and to be reasonably resistant to overfitting which are soft margin that prohibits fitting every noise point. As a representative margin-based classifier with a distinct inductive bias from trees or probabilistic models, we have incorporated SVM into our methodology. Because there are significantly fewer minority points that are "pushing back" on the decision boundary, class imbalance might cause the optimal hyperplane to move towards the minority side when SVM is applied on imbalanced data. In essence, because the majority class can dominate the objective, the unconstrained SVM solution may have a decision boundary that is very close to the minority instances (small margin on that side) if one class

significantly dominates the other. Data resampling or assigning different misclassification costs, a higher cost for incorrectly classifying a minority class—are two ways to deal with this. We use oversampling to choose the latter.

By oversampling the minority class, we provide the SVM more minority support vectors, which should lead to a more evenly distributed positioning of the separation hyperplane. The impact of the minority class points on the margin is increased by oversampling. Furthermore, SVM will be sensitive to any artificial or duplicate points that was created and falls on the edge of the minority distribution because it relies on support vectors. Techniques like SMOTE and ADASYN create minority points in the feature space to fill in the minority region and possibly offer a smoother margin. Depending on which fits the data better, we will use either a linear SVM or an RBF-kernel SVM to fit the model using the balanced training data. SVM's performance with the different sampling methods mentioned above will visualise how a margin-based approach benefits from the augmented minority data, especially in terms of improved minority recall. It is also worth noting that the SVM output is not a probabilistic estimate by default, but we can still evaluate its predictions with metrics such as F1-score and AUC that does not require a well-calibrated probability.

3.7 Evaluation

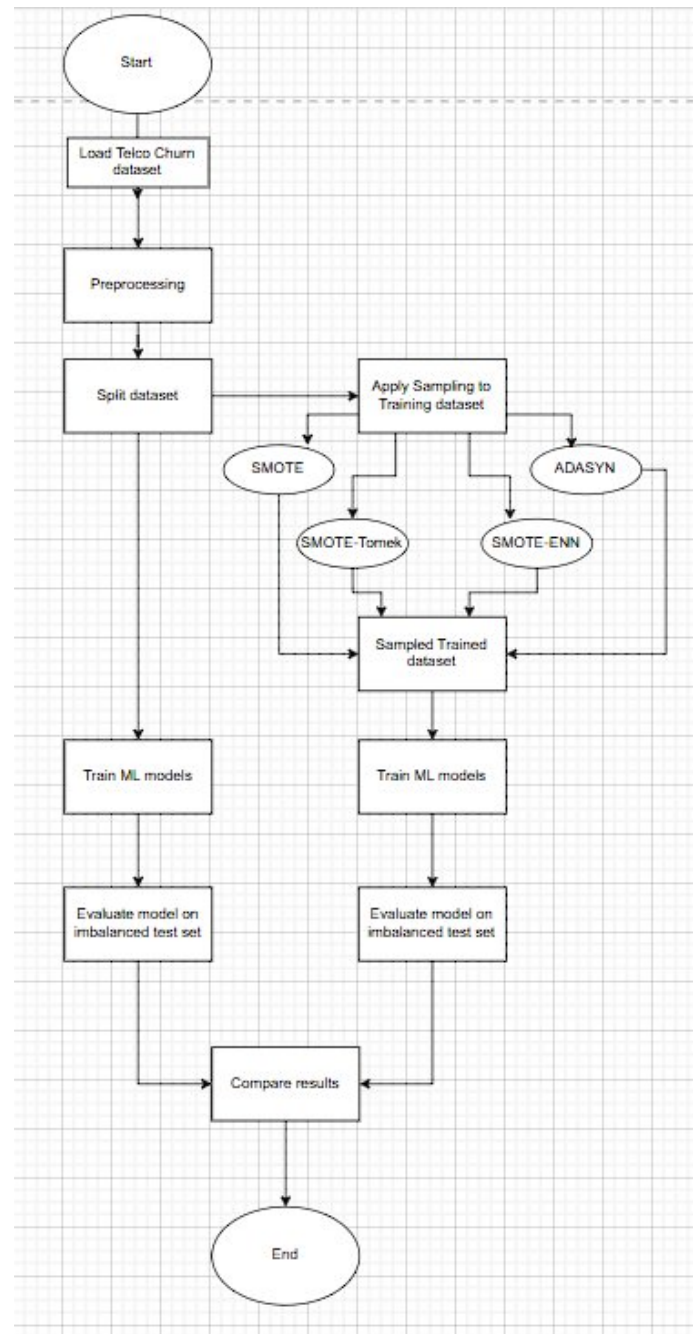


Fig 11. Overview of the training and evaluation workflow for churn prediction using resampling techniques.

We evaluated each of the training models on the independent test set which was not used in training or oversampling. This is due to the test set reflecting the original class imbalance, we use appropriate evaluation metrics that are beyond simple accuracy. If we just reply on accuracy in imbalanced settings, it can be misleading [39], for example, predicting all of the instances as majority may yield high accuracy, but it is considered a failed model for minority detection, and in this context of churn analysis minority detection is even more important.

Therefore, we must focus on metrics that will capture the minority class performance better: Precision, Recall (Sensitivity), F1-Score, and AUC (Area Under the ROC Curve) are all used to assess each model [39]. The Recall metric tells us how many of the predicted minority instances were correctly identified; Precision metric displays how many of the minority instances that were predicted are actually part of the minority class; F1-score is a metric offers a balanced measure that is maximized when both precision and recall are high because F1-score is the harmonic mean for both of it. AUC evaluates the trade-off between true positive rate and false positive rate across thresholds, which is useful for comparing the overall ranking performance irrespective of a specific threshold.

We should also consider the **Area Under the Precision-Call Curve (AUCPR)** if the class imbalance is too extreme, as it focuses on performance on the positive class. Using these metrics will provide a more robust evaluation under class imbalance, as recommended in literature [39]. We will then compare the scores of each model under each sampling method. Cross-validation can be employed on the training set to tune the model hyperparameters or to get an average performance estimate; in that case, the oversampling is applied within each fold to avoid any of the training data bleeding into validation.

Throughout the project, we have maintained consistent conditions for fairness. The random seed for oversampling and model initialization will be fixed where applicable SMOTE randomness, random forest's randomness to ensure reproducibility. If cross-validation is performed. A report on the mean and standard deviation of metrics across folds will be made. Training times will be observed too, as oversampling increases dataset size (except when combined with Tomek/ENN which might remove some instances) but given these are relatively small overheads for the dataset size, since all methods are computationally feasible.

According to the methodology, it aimed to address the class imbalance effectively and also to compare how each of the sampling strategies interacts with the different learning algorithms. The **combination of data level techniques such as oversampling, undersampling and algorithm-level choices** is expected to yield an optimal approach for the project's prediction task. In short, the methodology ensures that the classifiers are all trained on balanced, cleaner data, and are also evaluated with appropriate metrics, while aligning with best practices for imbalanced learning [39].

3.8 Tools, Technology and Deployment

To implement, evaluate, and visualize the machine learning models for churn prediction, a combination of cloud-based platforms, programming libraries, and data science tools are employed. The tools used (and to be used) in this capstone project are as follows:

Google Colab is a free cloud-based Jupyter notebook server offered by Google that does not require local hardware installation and configuration. It provides access to GPU and TPU acceleration, which is appropriate to computational intensive machine learning workloads. Google Colab has been considered in this work as the main development platform where the code, data preprocessing and model training, as well as the visualization of results were written, and the seamless integration with Python-based machine learning and data analysis libraries was also possible.

The main programming language of all the implementations of this project is **Python**, because it has a vast machine learning, data manipulation, and visualization ecosystem. It has a simple and readable syntax, which allows quick prototyping and experimentation, and its massive user base means that there is a well-maintained and broadly-used library of libraries that researchers need to develop data-driven research with.

One of the most popular **machine learning libraries** in Python is **scikit-learn** and it was used to create, train, and assess the predictive models of this paper. It facilitates the use of major classifiers like Decision Tree, Logistic regression, Rand Forest and Support Vector machine (SVM). Moreover, scikit-learn also has necessary tools to split the data, sample it stratified, and assess its performance by such measures as accuracy, precision, recall, F1-score, and Area Under the Curve (AUC).

Specialized Python package **imbalance-learn (imblearn)** was created to tackle the issue of class imbalance that is typical of real-world data. It was applied in this project to apply different techniques of oversampling such as SMOTE, SMOTE-Tomek, SMOTE-ENN and ADASYN. The approaches can be used to improve the representation of minority classes which is very important to improve the model performance in churn prediction tasks.

Pandas was also applied in data analysis, which involved data loading, data preprocessing, filtering, and exploratory analysis. Its robust data representations, including DataFrames, allow

working with tabular data efficiently and structurally and is, therefore, especially helpful to process and analyze telecom churn datasets.

NumPy is a **Python library** that is a basic library used to carry out work with numbers and mathematics effectively. It is important in the management of the computations that have been vectorized and in the management of the intermediate data at the time of sampling, feature processing as well as the training of the model to enhance the efficiency of calculations and consistency.

The **visualization of the results of data distributions and model performance** was done using **matplotlib** and **seaborn** to plot and make charts. These libraries assist in the graphic visualization of statistics and assist in providing analytical results in an understandable and effective way, which facilitates the interpretation and comparative analysis of the various models and sampling methods.

4.0 Results

This section presents the results that was obtained from evaluating four machine learning models Decision Tree (DT), Logistic Regression (LR), Random Forest (RF), and Support Vector Machine (SVM) across a range of different sampling techniques designed to address class imbalance in the telecom churn dataset. The performance of each model was assessed using two primary metrics: F1-score and AUC (Area Under the ROC Curve). These metrics were chosen because they provide a balanced view of minority-class predictive power and overall discrimination capability.

4.1 Overall Performance Summary

Across all of the testings, oversampling techniques have produced the most notable improvements over the baseline results for most models. The combined results were all organized into a consolidated performance table that contains all of the sample model combinations, then sorted by model, AUC, and F1-score. A grouped bar chart visualising F1-score and AUC by sampling techniques for each model highlights clear performance differences, while heatmap provides an intuitive comparison of sampling influence across all models.

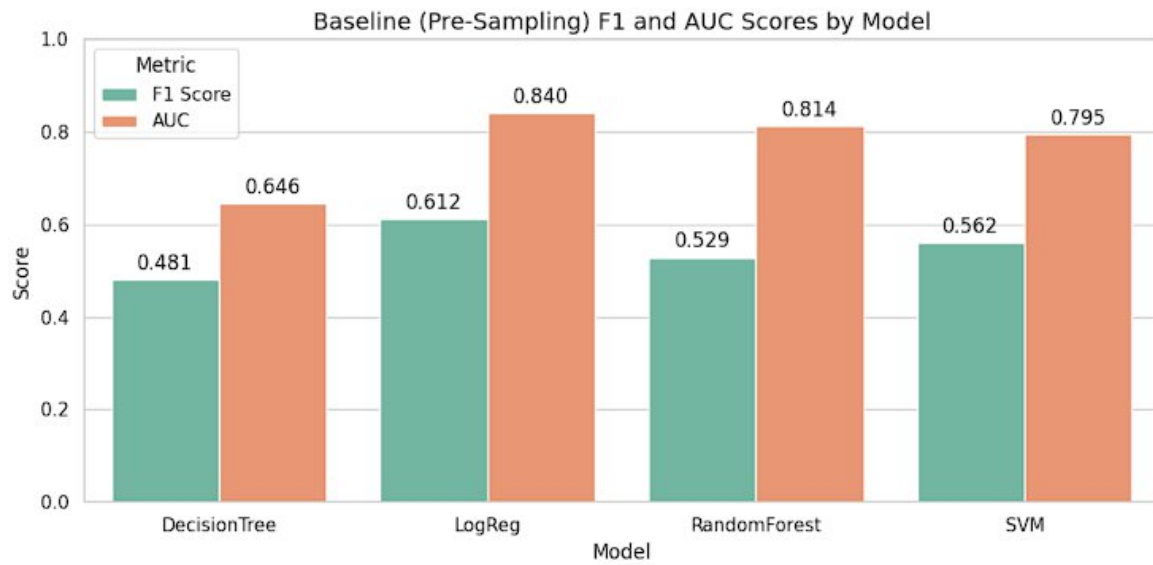


Fig 12. Baseline Metrics

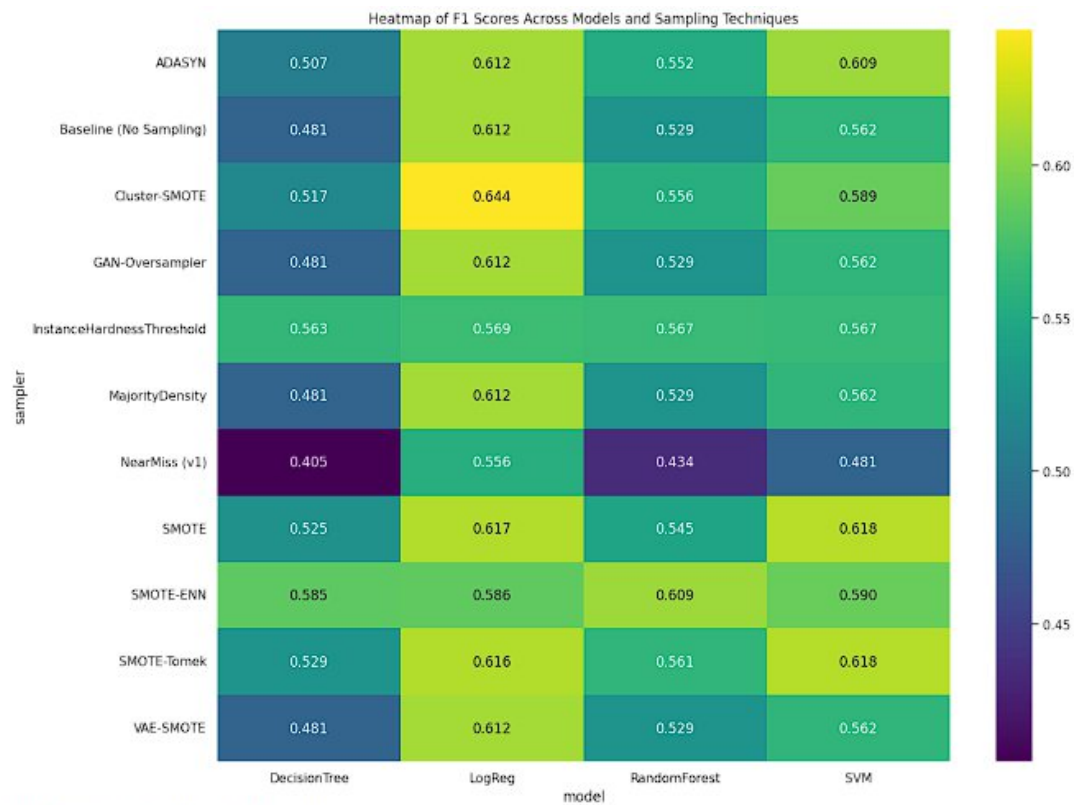


Fig 13. Heatmap of F1-Score

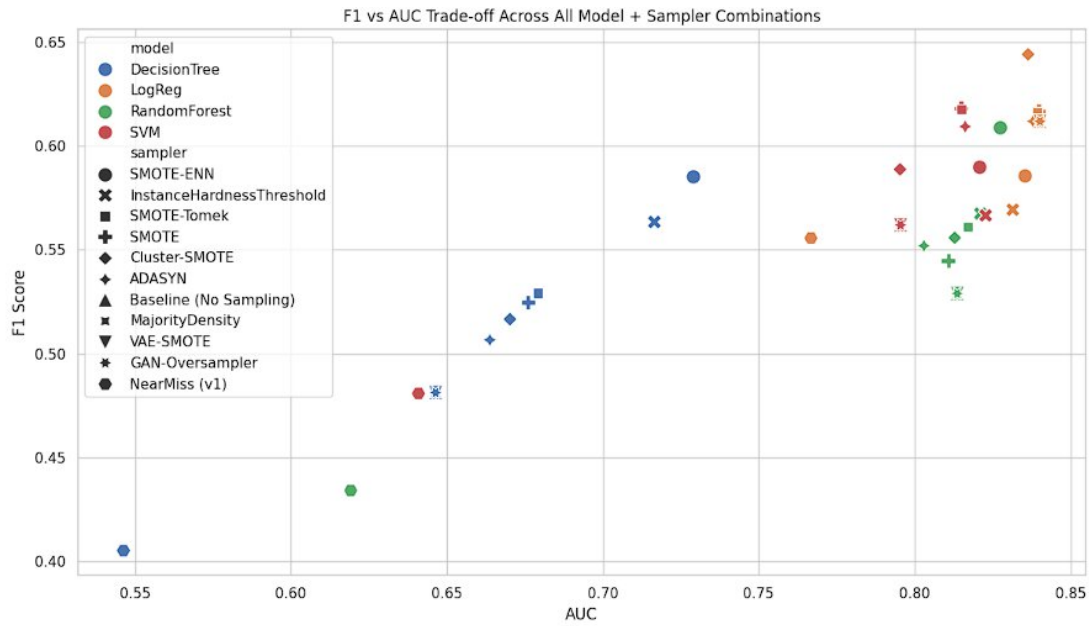


Fig 14. Scatterplot for comparison of F1-score and AUC

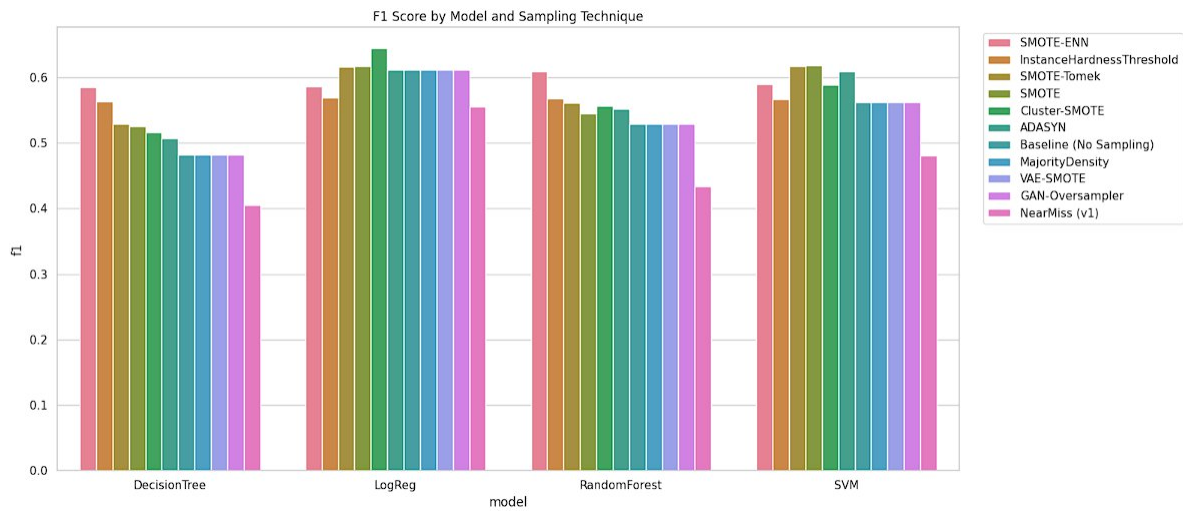


Fig 15. Overall F1 results

The figure above shows that Logistic Regression has produced the most optimal and consistent results after its data is sampled. The F1-score is the key evaluation metric in imbalanced classification because it balances both precision and recall of the minority class. When a sampling technique is effective, the F1-score will increase, indicating that the model can detect more minority samples while keeping misclassifications low. If the F1-score decreases or shows no improvement, the sampling method may have introduced noise, caused overfitting, or failed to improve minority class learning. Therefore, F1-score directly reflects whether a sampling technique has produced better, worse, or unchanged performance for minority class prediction.

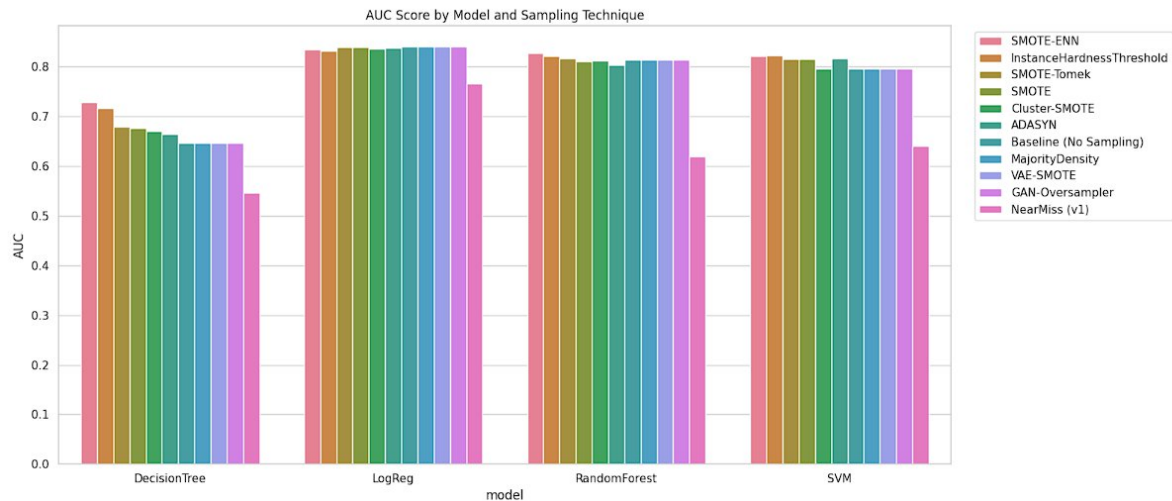


Fig 16. Overall AUC

AUC is another important indicator of whether a sampling technique improves the classifier's discriminatory ability in an imbalanced dataset. When sampling increases the AUC score, it usually means the model is better at ranking minority-class instances higher than majority ones across all thresholds. This reflects true improvement in separability rather than just accuracy at a fixed threshold. In contrast, a decrease in AUC indicates that the sampling method introduced noise or removed important samples, reducing the model's ability to correctly distinguish positive and negative classes. Based on the AUC results across Decision Tree, Logistic Regression, Random Forest, and SVM, methods such as SMOTE, SMOTE-Tomek, Cluster-SMOTE, VAE-SMOTE, and GAN-Oversampling consistently improved or stabilised AUC, while NearMiss significantly reduced AUC due to aggressive under-sampling.

Overall, the results have indicated that data imbalance handling does substantially affect model performances, and the effectiveness of each sampling technique varies depending on the underlying algorithm

SVM showed strong and consistent improvements under oversampling. As seen from Fig 15 SMOTE has achieved the highest F1-score of **0.618**, closely followed by SMOTE-Tomek (**0.617**). The top-performing methods for SVM produced both improved F1 and stable, high AUC values around the **0.815–0.822** range.

Cluster-SMOTE produced moderate improvements, while baseline performance was slightly lower (F1 = **0.562**). Generative sampling again did not produce measurable gains. Once again

NearMiss (v1) produced significantly worse results for SVM, with both F1 and AUC dropping sharply.

4.2 F1-AUC Trade-off Patterns

The F1–AUC scatter plot provides a holistic view of the performance landscape. Techniques that lie in the upper-right quadrant such as SMOTE-ENN for tree-based models, Cluster-SMOTE for Logistic Regression, and SMOTE for SVM represent the most balanced improvements, a trend that aligns with findings in previous studies showing that hybrid oversampling methods tend to preserve structure while enhancing minority representation [40], [41]. Conversely, NearMiss consistently occupies the lower-left region, confirming its negative impact across all models, as aggressive undersampling often results in the removal of informative majority-class examples that are crucial for discriminative learning [42].

4.3 Key Observations from Results

Using the experimental results in the table, it is possible to note several consistent patterns of performance when comparing the sampled techniques and machine learning models used. In general, the hybrid sampling techniques, especially SMOTE-ENN, SMOTE-Tomek and Instance Hardness Threshold (IHT) methods always beats the simple oversampling techniques. This explains the significance of using oversampling with noise or borderline case dropping in the face of class imbalance in telecom churn data.

Logistic Regression was selected as the most stable and consistent classifier in regard to performance improvements when compared to almost all sampling techniques. This makes it quite strong and stable particularly in the case of imbalanced learning where constant generalization is very imperative. Decision Tree models on the other hand, were observed to be extremely sensitive to data noise but significantly advantaged by noise-minimising samplers like SMOTE-ENN and SMOTE-Tomek that acted to enhance predictive performance by alleviating the effects of mislabeled or uncertain samples [40].

Also, the findings suggest a good fit between SMOTE and Support Vector Machine (SVM) in which the oversampling greatly improved the performance of SVM. This can be observed in accordance with previous results that a more balanced representation of minority samples can be used to advantage margin-based classifiers during training. On the other hand, NearMiss (v1) continued to yield the most poor results of all the models tested, which proves that the

aggressive undersampling schemes are capable of eliminating the important minority-class data and adversely impacting the churn prediction accuracy.

All these observations together prove that the sampling technique selection is a more important element than the model complexity itself. Although more sophisticated classifiers are possible, suitable combinations of sampling strategies and traditional machine learning algorithms like Logistic Regression, Decision Tree, Random Forest, and SVM have a higher guarantee of effective execution as well as stability and functionality in actual churn prediction endeavors.

5.0 Discussion

This section will provide an in-depth interpretation of the findings presented earlier. The purpose is to help explain why certain sampling techniques and models performed better, how the visual patterns relate to theoretical expectations, and what these results imply for churn prediction in imbalanced datasets. The discussion integrates observations from the grouped bar charts, heatmaps, ranking plots, and F1–AUC scatter visualizations.

5.1 Impact of Sampling techniques

Across all models, the visualizations indicate that sampling techniques substantially influence both F1 and AUC. The grouped bar chart and heatmap reveal that oversampling consistently improves minority-class detection compared to baseline performance, especially for Logistic Regression, Random Forest, and SVM. This aligns with prior research showing that class imbalance severely degrades classification performance and that synthetic oversampling improves model ability to capture minority-class structure[40], [41].

However, not all sampling techniques perform equally. Hybrid methods such as SMOTE-ENN and SMOTE-Tomek generally outperform simpler oversamplers, whereas aggressive undersampling (e.g., NearMiss v1) consistently results in the weakest performance. These patterns confirm findings from the imbalanced learning literature, which states that hybrid oversampling with noise reduction tends to improve class separability, whereas undersampling risks discarding valuable information required for reliable decision boundaries [42].

5.2 Performance of Tree-Based Models

Decision Tree

The Decision Tree model has shown the greatest sensitivity to sampling choices. SMOTE-ENN have achieved the highest F1-score of **0.585**, followed by IHT and SMOTE-Tomek. The ranking charts have illustrated that hybrid approaches had dominated the upper tier of results, whereas simple oversamplers such as SMOTE, ADASYN and generative methods like VAE-SMOTE, GAN-Oversampler cluster closer to the baseline.

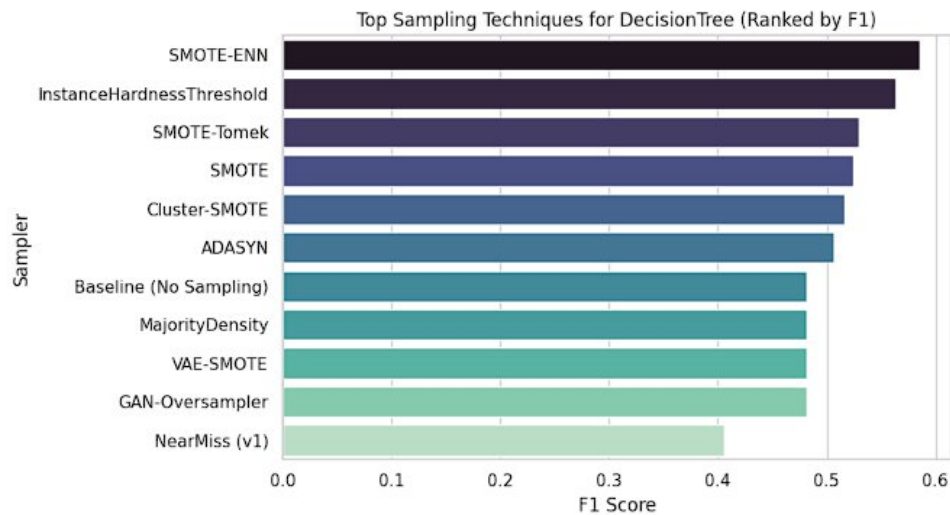


Fig 17. Top sampling techniques for Decision Tree

This behavior is very much expected because Decision Trees are highly sensitive to noisy or borderline points. Hybrid techniques like SMOTE-ENN remove mislabeled or ambiguous samples, leading to cleaner splits and improved generalization [40]. On the other hand, NearMiss v1 dramatically underperformed, which solidifies the observations that undersampling often deletes informative majority examples which in turn weakens the model's ability to construct meaningful decisions [42].

Random Forest

Random forest have showed a similar pattern, with SMOTE-ENN again emerging as the strongest method. Its ensemble nature makes it more resilient to noise than a single Decision Tree, but it still benefits notably from hybrid preprocessing. The scatter plot shows Random Forest results distributed in the upper-middle region, with hybrid methods yielding the best balance of F1 and AUC.

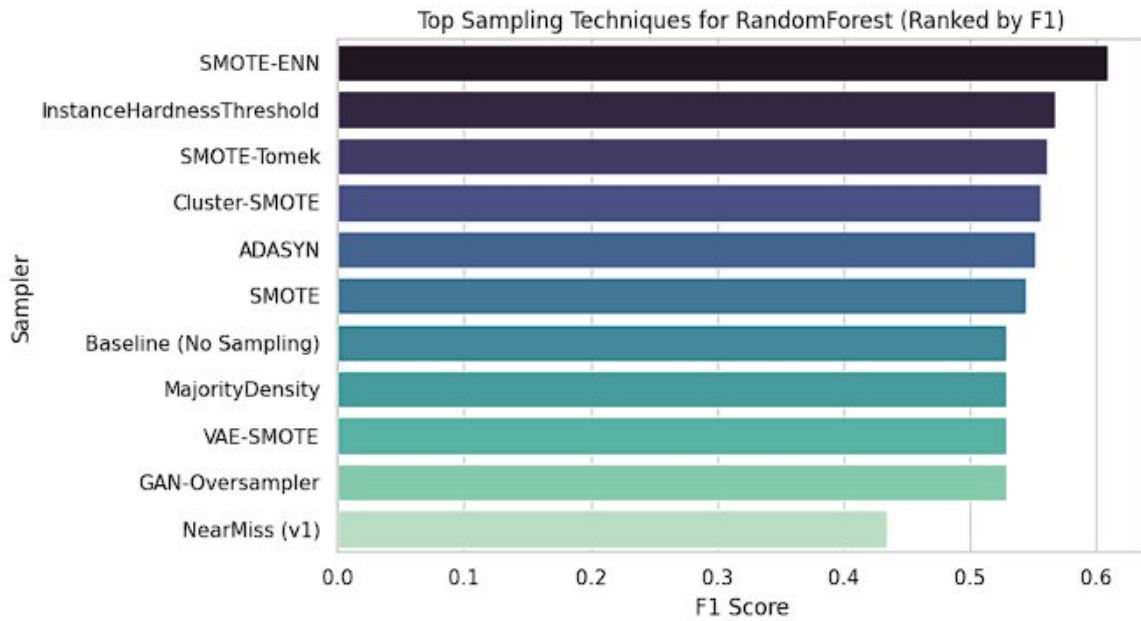


Fig 18. Top sampling techniques for Random Forest

Generative oversampling methods again showed no improvement over baseline, an outcome consistent with studies showing that GAN-based tabular oversampling often struggles to generate realistic feature combinations, especially in structured datasets like telecom churn [43], [44].

5.3 Performance of Logistic Regression

Logistic Regression demonstrated the most stable and consistent improvements across all sampling techniques. The most that achieved the highest F1-score was Cluster-SMOTE with also a competitive AUC of **0.836**, as highlighted in the heatmap and ranking plot. This aligns with prior findings that Cluster-based synthetic sampling preserves structural relationships between minority and majority clusters, resulting in smoother linear decision boundaries [41], [45].

Interestingly, the baseline AUC for Logistic Regression remained a high **0.84**, indicating that LR already captures significant discriminatory information. However, its F1 score increased meaningfully after oversampling due to improved recall of churners. This illustrates that LR benefits from additional minority examples without suffering from severe boundary distortion—an observation also reported in earlier work [45]

5.4 Performance of Support Vector Machine

SVM responded strongly to the traditional oversampling approaches, with SMOTE and SMOTE-Tomek achieving the highest F1-scores of **0.618** respectively. The ranking plot have

showed both methods outperforming the hybrid approaches, which differs from the behaviour of tree-based models. This aligns with theoretical expectations: SVM constructs a maximal-margin hyperplane, and the synthetic points produced by SMOTE expand the minority-class region, making the margin more balanced [25], [40].

Cluster-SMOTE and IHT provided moderate improvements, while generative oversamplers again showed little gain. NearMiss v1 performed the worst, consistent with its tendency to remove informative support vectors, which are critical for defining SVM decision boundaries [42].

5.5 Insights from the F1-AUC Scatter Plot

The F1-AUC scatter plot as shown in Fig 14 offers a holistic perspective of the model and sampler combinations. There are three major regions that are observed here:

- 1) Upper right quadrant (best overall results)

This area contains SMOTE-ENN (tree models), Cluster-SMOTE (LR), and SMOTE/SMOTE-Tomek for SVM. These methods have delivered optimal balance between minority recall (F1) and overall discrimination (AUC), consistent with prior work on hybrid oversampling effectiveness [40], [41].

- 2) Central Clusters (moderate performance)

This area includes methods such as ADASYN and IHT, which represents moderate improvements without boundary refinements.

- 3) Lower left quadrant (poor performance)

This area is mainly dominated by NearMiss v1 and some of the baseline results, demonstrating inadequate handling of the minority class representation.

The scatter plot have also shown that VAE-SMOTE and GAN-Oversampler cluster near the baseline performance. This pattern is consistent throughout the project which shows us that generative models underperform on high dimensional tabular data where accurate distribution modelling is difficult [43], [44].

5.6 Practical Implications for Churn Prediction

From an organizational perspective, churn prediction requires maximising recall of minority churners while retaining high discrimination.

Overall, the findings indicate that the effectiveness of churn prediction models is strongly influenced by the choice of sampling technique. Tree-based models benefit most from noise-reducing hybrid approaches such as SMOTE-ENN, which help mitigate the impact of

misabeled or ambiguous instances while improving class balance. Logistic Regression demonstrates strong stability across sampling strategies and achieves its best performance when combined with Cluster-SMOTE, reinforcing its suitability for interpretable and business-oriented churn prediction applications. Support Vector Machine (SVM) models perform optimally with SMOTE and SMOTE-Tomek, where balanced class distributions help preserve smooth and well-defined decision margins. In contrast, aggressive undersampling techniques such as NearMiss v1 consistently degrade predictive performance and should be avoided in churn analysis due to the loss of valuable minority-class information. These findings have provided us with a clear roadmap for data scientists and telecom operators when selecting preprocessing strategies to enhance churn prediction.

5.7 Summary of Discussion

Overall, the project results have demonstrated that the sampling technique selection plays a critical role in model performance for imbalanced telecom churn classification. Hybrid oversamplers consistently outperform simpler techniques, Logistic Regression delivers highly stable results, and generative oversamplers do not provide meaningful improvements. These outcomes align strongly with established imbalanced learning literature and provide clear guidance for practical deployment.

6.0 Conclusion

6.1 Overall Conclusion

This project has examined the impact of multiple sampling techniques on the performance of the four machine learning models which are Decision Tree, Logistic Regression, Random Forest, and Support Vector Machine in predicting telecom customer churn under significant class imbalance. Through a comprehensive evaluation using F1-score and AUC, supported by visual analyses such as grouped bar charts, heatmaps, ranking plots, and F1–AUC scatter diagrams, several meaningful insights were obtained.

The results have consistently demonstrated that sampling strategy selection has a substantial effect on predictive performance. Hybrid oversampling methods, particularly SMOTE-ENN, provided the most significant improvements for tree-based models by combining synthetic sample generation with effective noise reduction. Logistic Regression exhibited the highest stability and achieved its strongest performance with Cluster-SMOTE, suggesting that

structured, cluster-preserving synthetic samples enhance linear decision-making boundaries. For SVM, traditional oversampling techniques such as SMOTE and SMOTE-Tomek yielded the best results by expanding the minority representation in margin-relevant regions. In contrast, undersampling methods especially NearMiss v1 consistently degraded performance across all models due to excessive removal of informative majority-class instances.

Another notable finding is that generative oversampling approaches such as VAE-SMOTE and GAN-based sampling did not offer substantial performance gains. These techniques produced results like baseline performance, reflecting the challenges generative models face when synthesizing realistic tabular data structures for complex domains like telecom churn.

Overall, the study confirms that model behaviour under class imbalance is closely tied to the choice of sampling technique, and no single method is universally optimal across all algorithms. However, clear patterns emerge which are hybrid oversamplers are most effective for tree-based models, cluster-based oversampling benefits Logistic Regression, and margin-based models like SVM align best with SMOTE-based techniques.

These findings offer practical value for organisations seeking to improve churn prediction systems. By selecting sampler and model combinations that align with the observed performance characteristics, telecom operators can enhance the accuracy of churn identification, support targeted retention strategies, and reduce customer loss.

The results also highlight potential areas for future work, such as integrating cost-sensitive learning, testing more advanced generative models, and deploying real-time adaptive sampling strategies. In conclusion, effective handling of class imbalance is a critical step in developing reliable churn prediction models, and the insights from this study contribute to a more informed approach in selecting and applying sampling techniques within machine learning workflows. Despite the encouraging results achieved in this study, several limitations remain, which provide opportunities for future research.

6.2 Limitations

Although this study provided many meaningful insights into the impact of sampling techniques on churn prediction performance, although there are several limitations that I think should be acknowledged to contextualise the findings and guide future researches on this topic.

6.2.1 Data Constraints

The analysis was conducted only on a single telecom churn dataset, which may greatly limit the generalisability of the findings. The dataset also contains a substantial imbalance between churn and non-churn classes, and while sampling techniques helped mitigate this issue, some imbalance-related challenges may persist. Additionally, the dataset does not capture temporal or behavioral patterns such as time-series usage trends, which could provide richer predictive signals.

6.2.2 Model Scope

This study focused on four classical machine learning models—Decision Tree, Logistic Regression, Random Forest, and SVM. While these models are widely used in churn prediction, the exclusion of more advanced methods such as gradient boosting, ensemble meta-learners, or deep learning models may limit the breadth of comparative insights.

6.2.3 Sampling Method Constraints

The performance of generative models such as VAE-SMOTE and GAN-based oversamplers was limited even though a wide range of oversampling and undersampling techniques were tested in this project. This may be due to possibly the difficulty generative models will face when learning realistic distributions from high-dimensional tabular data. Additionally, some sampling techniques introduce synthetic noise, which may affect model stability, especially for noise-sensitive algorithms such as Decision Trees.

6.2.4 Hyperparameter and Pipeline Limitations

Model hyperparameters and sampling ratios were not exhaustively optimised due to computational constraints. More extensive tuning could potentially yield higher performance. Moreover, adding variations in encoder choices, preprocessing steps, or pipeline configurations may influence the model and sampler dynamics.

6.2.5 Evaluation Metric Limitations

Although F1-score and AUC provide strong insight into the model's performances, other metrics such as Precision-Recall AUC, Lift, Gain, or business impact analysis have not been explored here. These metrics can surely provide a more comprehensive evaluation of real world churn prediction effectiveness.

6.3 Future Work

6.3.1 Expanded model Exploration

Future studies should incorporate advanced models such as XGBoost, LightGBM, CatBoost, and deep learning architectures like TabNet or transformer-based models for tabular data. These models may capture more complex customer patterns and interact differently with sampling techniques.

6.3.2 Improved Generative Oversampling Methods

Further studies into advanced generative models such as CTGAN, TVAE, or diffusion-based tabular synthesizers could help address the limitations of traditional GAN and VAE oversampling. These methods may be able to produce more realistic synthetic samples and improve performance in structured datasets.

6.3.3 Cost-Sensitive and Ensemble-Based Approaches

In addition to data-level sampling, algorithm-level imbalance handling such as cost-sensitive learning, Balanced Random Forest, RUSBoost, and EasyEnsemble should be explored. These techniques may be able to offer a stronger alternative to sampling by directly penalizing misclassification of minority-class churners.

6.3.4 Hyperparameter Optimization and Sampling Intensity Tuning

Future work should include automated search strategies like grid search, random search, or Bayesian optimization to refine a model's hyperparameter and optimize sampling ratios. Fine-tuning sampling intensity may further improve performance and reduce synthetic noise.

6.3.5 Integration of Explainability and Business Metrics

Explainable AI (XAI) techniques such as SHAP or LIME could be integrated to interpret model predictions and identify key drivers of churn. Additionally, business-focused evaluation

methods such as customer lifetime value, retention cost analysis should be included to align model performance with operational goals.

6.3.6 Use of temporal and Behavioral Data

Future research should consider using longitudinal datasets that capture time-based customer behaviour. Incorporating trends such as monthly usage decline, payment delays, or service complaints could enhance prediction accuracy and provide more actionable insights.

Reference

- [1] T. Tang, "Class Imbalance Strategies — A Visual Guide with Code," TDS Archive. Accessed: Nov. 29, 2025. [Online]. Available: <https://medium.com/data-science/class-imbalance-strategies-a-visual-guide-with-code-8bc8fae71e1a>
- [2] I. Ullah, B. Raza, A. K. Malik, M. Imran, S. U. Islam, and S. W. Kim, "A Churn Prediction Model Using Random Forest: Analysis of Machine Learning Techniques for Churn Prediction and Factor Identification in Telecom Sector," *IEEE Access*, vol. 7, pp. 60134–60149, 2019, doi: 10.1109/ACCESS.2019.2914999.
- [3] K. Dahiya and S. Bhatia, "Customer churn analysis in telecom industry," in *2015 4th International Conference on Reliability, Infocom Technologies and Optimization (ICRITO) (Trends and Future Directions)*, Noida, India: IEEE, Sept. 2015, pp. 1–6. doi: 10.1109/ICRITO.2015.7359318.
- [4] A. A. Ahmed and D. Maheswari, "A review and analysis of churn prediction methods for customer retention in telecom industries," presented at the 2017 International Conference on Advanced Computing and Communication Systems (ICACCS), 2017, pp. 1–6. doi: 10.1109/ICACCS.2017.8014664.
- [5] A. Bhatnagar and S. Srivastava, "Customer Churn Prediction: A Machine Learning Approach with Data Balancing for Telecom Industry," *Int. J. Comput.*, pp. 9–18, Mar. 2025, doi: 10.47839/ijc.24.1.3873.
- [6] S. S. Poudel, S. Pokharel, and M. Timilsina, "Explaining customer churn prediction in telecom industry using tabular machine learning models," *Mach. Learn. Appl.*, vol. 17, p. 100567, Sept. 2024, doi: 10.1016/j.mlwa.2024.100567.
- [7] T. A. R. Akbar and C. Apriono, "Machine Learning Predictive Models Analysis on Telecommunications Service Churn Rate," *Green Intell. Syst. Appl.*, vol. 3, no. 1, Art. no. 1, June 2023, doi: 10.53623/gisa.v3i1.249.
- [8] B. Prabadevi, R. Shalini, and B. R. Kavitha, "Customer churning analysis using machine learning algorithms," *Int. J. Intell. Netw.*, vol. 4, pp. 145–154, 2023, doi: 10.1016/j.ijin.2023.05.005.
- [9] M. Chajia and E. Nfaoui, "Customer Churn Prediction Approach Based on LLM Embeddings and Logistic Regression," *Future Internet*, vol. 16, p. 453, Dec. 2024, doi: 10.3390/fi16120453.
- [10] M. D. N. Alif and N. F. Fahrudin, "Performance Analysis of Oversampling and Undersampling on Telco Churn Data Using Naive Bayes, SVM And Random Forest

- Methods,” *E3S Web Conf.*, vol. 484, p. 02004, 2024, doi: 10.1051/e3sconf/202448402004.
- [11] A. Sikri, R. Jameel, S. M. Idrees, and H. Kaur, “Enhancing customer retention in telecom industry with machine learning driven churn prediction,” *Sci. Rep.*, vol. 14, no. 1, p. 13097, June 2024, doi: 10.1038/s41598-024-63750-0.
- [12] M. Imani, Z. Ghaderpour, M. Joudaki, and A. Beikmohammadi, *The Impact of SMOTE and ADASYN on Random Forest and Advanced Gradient Boosting Techniques in Telecom Customer Churn Prediction*. 2024. doi: 10.20944/preprints202403.0213.v2.
- [13] S. Pun, S. Thapa, and S. Timilsina, “Customer Churn Prediction Using ADASYN Sampling Technique and Ensemble Model,” vol. 6, Aug. 2021.
- [14] Kimura and T, “Customer churn prediction with hybrid resampling and ensemble learning,” *J. Manag. Inf. Decis. Sci.*, vol. 25, no. 1, pp. 1–113, Dec. 2021.
- [15] M. Imani and H. R. Arabnia, “Hyperparameter Optimization and Combined Data Sampling Techniques in Machine Learning for Customer Churn Prediction: A Comparative Analysis,” *Technologies*, vol. 11, no. 6, p. 167, Dec. 2023, doi: 10.3390/technologies11060167.
- [16] R. Suguna, J. Suriya Prakash, H. Aditya Pai, T. R. Mahesh, V. Vinoth Kumar, and T. E. Yimer, “Mitigating class imbalance in churn prediction with ensemble methods and SMOTE,” *Sci. Rep.*, vol. 15, no. 1, p. 16256, May 2025, doi: 10.1038/s41598-025-01031-0.
- [17] G. Douzas, F. Bacao, and F. Last, “Improving imbalanced learning through a heuristic oversampling method based on k-means and SMOTE,” *Inf. Sci.*, vol. 465, pp. 1–20, Oct. 2018, doi: 10.1016/j.ins.2018.06.056.
- [18] Y. Lin and Z. Wang, “Class imbalance handling via deep generative models: A variational autoencoder based approach,” *Expert Syst. Appl.*, vol. 181, p. 115162, 2021, doi: 10.1016/j.eswa.2021.115162.
- [19] A. Faraji, J. El Khatib, M. A. Maqsood, and B. Ravi, “A hybrid approach for customer churn prediction using CTGAN and Edited Nearest Neighbor,” *J. Big Data*, vol. 11, no. 1, pp. 1–21, 2024, doi: 10.1186/s40537-024-00854-y.
- [20] Y. Kubat and S. Matwin, “Addressing the curse of imbalanced training sets: One-sided selection,” in *Proceedings of the 14th International Conference on Machine Learning (ICML)*, 1997, pp. 179–186. [Online]. Available: <https://sci2s.ugr.es/keel/pdf/algorithm/articulo/kubat1997.pdf>

- [21] A. AlShourbaji and N. Helian, "Churn Prediction for Telecommunication Research Papers - Academia.edu." Accessed: Nov. 29, 2025. [Online]. Available: https://www.academia.edu/Documents/in/Churn_Prediction_for_Telecommunication?f_ri=21395
- [22] M. Imani, M. Joudaki, A. Beikmohammadi, and H. Arabnia, *Customer Churn Prediction: A Systematic Review of Recent Advances, Trends, and Challenges in Machine Learning and Deep Learning*. 2025. doi: 10.20944/preprints202503.1969.v1.
- [23] "Crisp DM methodology," Smart Vision Europe. Accessed: Nov. 29, 2025. [Online]. Available: <https://www.sv-europe.com/crisp-dm-methodology/>
- [24] "1.10. Decision Trees — scikit-learn 1.7.2 documentation." Accessed: Nov. 29, 2025. [Online]. Available: <https://scikit-learn.org/stable/modules/tree.html>
- [25] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic Minority Over-sampling Technique," *J. Artif. Intell. Res.*, vol. 16, pp. 321–357, June 2002, doi: 10.1613/jair.953.
- [26] I. Tomek, "Two Modifications of CNN," *IEEE Trans. Syst. Man Cybern.*, vol. SMC-6, no. 11, pp. 769–772, Nov. 1976, doi: 10.1109/TSMC.1976.4309452.
- [27] D. L. Wilson, "Asymptotic Properties of Nearest Neighbor Rules Using Edited Data," vol. SMC-2, no. 3, pp. 408–421, 1972.
- [28] H. He, Y. Bai, E. A. Garcia, and S. Li, "ADASYN: Adaptive synthetic sampling approach for imbalanced learning," in *2008 IEEE International Joint Conference on Neural Networks (IEEE World Congress on Computational Intelligence)*, June 2008, pp. 1322–1328. doi: 10.1109/IJCNN.2008.4633969.
- [29] "Decision tree learning," *Wikipedia*. Oct. 06, 2025. Accessed: Nov. 29, 2025. [Online]. Available: https://en.wikipedia.org/w/index.php?title=Decision_tree_learning&oldid=1315457088
- [30] "Logistic Regression – Multi-Class Classification," Campus SDH. Accessed: July 24, 2025. [Online]. Available: <https://campus.sdh.gba.gob.ar/logistic-regression-multi-class-classification>
- [31] L. Breiman, "Random Forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, Oct. 2001, doi: 10.1023/A:1010933404324.
- [32] "What Is Logistic Regression? | IBM." Accessed: Dec. 17, 2025. [Online]. Available: <https://www.ibm.com/think/topics/logistic-regression>
- [33] "Sigmoid Function - GeeksforGeeks." Accessed: Nov. 29, 2025. [Online]. Available: <https://www.geeksforgeeks.org/machine-learning/derivative-of-the-sigmoid-function/>

- [34] “Random forest,” *Wikipedia*. Sept. 15, 2025. Accessed: Nov. 29, 2025. [Online]. Available:
https://en.wikipedia.org/w/index.php?title=Random_forest&oldid=1311485174
- [35] “What Is Random Forest? | IBM.” Accessed: Nov. 29, 2025. [Online]. Available:
<https://www.ibm.com/think/topics/random-forest>
- [36] “Support vector machine,” *Wikipedia*. Oct. 23, 2025. Accessed: Nov. 29, 2025. [Online]. Available:
https://en.wikipedia.org/w/index.php?title=Support_vector_machine&oldid=1318343861
- [37] N. V. Otten, “Support Vector Machines (SVM) In Machine Learning Made Simple & How To Tutorial,” Spot Intelligence. Accessed: Nov. 29, 2025. [Online]. Available:
<https://spotintelligence.com/2024/05/06/support-vector-machines-svm/>
- [38] F. Pedregosa, “SVM: Maximum margin separating hyperplane — scikit-learn 1.7.2 documentation.” Accessed: Nov. 29, 2025. [Online]. Available: https://scikit-learn.org/stable/auto_examples/svm/plot_separating_hyperplane.html
- [39] H. He and E. A. Garcia, “Learning from Imbalanced Data,” *Knowl. Data Eng. IEEE Trans. On*, vol. 21, pp. 1263–1284, Oct. 2009, doi: 10.1109/TKDE.2008.239.
- [40] G. Batista, R. Prati, and M.-C. Monard, “A Study of the Behavior of Several Methods for Balancing machine Learning Training Data,” *SIGKDD Explor.*, vol. 6, pp. 20–29, June 2004, doi: 10.1145/1007730.1007735.
- [41] V. García, J. S. Sánchez, and R. A. Mollineda, “On the effectiveness of preprocessing methods when dealing with different levels of class imbalance,” *Knowl.-Based Syst.*, vol. 25, no. 1, pp. 13–21, Feb. 2012, doi: 10.1016/j.knosys.2011.06.013.
- [42] X.-Y. Liu, J. Wu, and Z.-H. Zhou, “Exploratory Undersampling for Class-Imbalance Learning,” *IEEE Trans. Syst. Man Cybern. Part B Cybern.*, vol. 39, no. 2, pp. 539–550, Apr. 2009, doi: 10.1109/TSMCB.2008.2007853.
- [43] X. Zhang, F. Li, and X. Hu, “Enhancing Tabular Data Generation With Dual-Scale Noise Modeling,” *IEEE Access*, vol. 13, pp. 48643–48655, 2025, doi: 10.1109/ACCESS.2025.3551594.
- [44] D. S. B. von Ischmiddey, “Data Augmentation for tabular data,” Data Science Blog von Ischmiddey. Accessed: Dec. 09, 2025. [Online]. Available:
https://Ischmiddey.github.io/fastpages_/2021/03/14/tabular-data-variational-autoencoder.html

- [45] C. Bunkhumpornpat, K. Sinapiromsaran, and C. Lursinsap, “Safe-Level-SMOTE: Safe-Level-Synthetic Minority Over-Sampling TEchnique for Handling the Class Imbalanced Problem,” in *Advances in Knowledge Discovery and Data Mining*, T. Theeramunkong, B. Kijssirikul, N. Cercone, and T.-B. Ho, Eds., Berlin, Heidelberg: Springer, 2009, pp. 475–482. doi: 10.1007/978-3-642-01307-2_43.